

DESI DR2 Results IV: Alcock–Paczyński Measurements from the Lyman Alpha Forest and Cosmological Constraints

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We present Alcock–Paczynski (AP) measurements from the full shape of Lyman- α ($\text{Ly}\alpha$) forest correlation functions measured from the second data release (DR2) of the Dark Energy Spectroscopic Instrument (DESI). Our measurements include information from the $\text{Ly}\alpha$ forest three-dimensional auto-correlation and its cross-correlation with quasar positions. The analysis was validated using two types of mocks, several data splits, and extensive variations. We constrain the AP effect with 1% precision at an effective redshift $z_{\text{eff}} = 2.33$, which is twice as tight as the Baryon Acoustic Oscillation (BAO) constraint from the same data. When using the joint $\text{Ly}\alpha$ AP and BAO results, we measure the ratios $D_{\text{H}}(z_{\text{eff}})/r_{\text{d}} = 8.600 \pm 0.066$ (0.8%) and $D_{\text{M}}(z_{\text{eff}})/r_{\text{d}} = 39.32 \pm 0.33$ (0.8%), where D_{M} is the transverse comoving distance, D_{H} is the Hubble distance, and r_{d} is the size of the sound horizon at the drag epoch. Assuming ΛCDM , $\text{Ly}\alpha$ forest measurements combined with a big bang nucleosynthesis prior produce a constraint on the Hubble constant $H_0 = 66.5 \pm 1.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The $\text{Ly}\alpha$ AP result corresponds to a matter fraction constraint $\Omega_{\text{m}} = 0.325 \pm 0.018$ in ΛCDM , which is 1.4σ higher than DESI BAO. This impacts the DESI results relative to the Cosmic Microwave Background (CMB), slightly reducing their discrepancy from 2.4σ to 2.2σ . We present updated constraints on several extended models using the joint DESI DR2 BAO and $\text{Ly}\alpha$ forest full shape data, together with external data sets. When considering a time-evolving dark energy equation of state parametrized by w_0 and w_a , we find it is preferred over ΛCDM at 2.7σ for the combination of DESI and CMB data, and at 3.2σ when also including supernovae. With the new $\text{Ly}\alpha$ AP measurement, DESI provides its most precise anchor for the expansion history at $z > 1$ in the matter-dominated Universe.

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I. INTRODUCTION

The physics behind cosmic acceleration remains one of the most important problems in cosmology today, with progress requiring increasingly precise constraints of the cosmic expansion rate over a wide span of redshifts. Baryon acoustic oscillations (BAO) have long been central to studying the expansion history, with recent BAO measurements from the Dark Energy Spectroscopic Survey (DESI, [1–3]) providing intriguing hints that dark energy might not be described by a cosmological constant [4, 5]. DESI BAO measurements come from galaxy clustering at $z \lesssim 2$ and from the Lyman- α (Ly α) forest at $z \gtrsim 2$, with the two probes providing complementary constraints [e.g., 4, 6]. In this paper, we use the full shape of Ly α forest correlation functions measured from the second DESI data release (DR2) to significantly improve constraints on the cosmic expansion rate at $z > 2$. We also examine the cosmological implications of the new constraints when combined with DESI BAO results and with external probes.

The Ly α forest is the pattern of absorption in spectra of high-redshift quasars (QSO) and galaxies, caused by intervening neutral hydrogen in the intergalactic medium (IGM). The forest represents a unique tracer of large-scale structure (LSS), because it allows us to continuously map the density field along each line-of-sight [e.g., 7]. Over the last few decades, it has served as the primary tracer of LSS at high redshift ($z > 2$), a regime that has been difficult to access with large galaxy surveys thus far. The Ly α forest and its cross-correlation with high-redshift quasars have been used to measure BAO starting with the Baryon Oscillation Spectroscopic Survey (BOSS) [8] more than a decade ago [9, 10], continuing with the extended BOSS (eBOSS) survey [11, 12] and recently with DESI [13, 14].

Over the last few years, large scales in the Ly α forest correlation functions have also been used to extract cosmological information beyond BAO [15, 16]. These works leverage the Alcock–Paczyński (AP) effect [17], which imprints a characteristic anisotropy due to differences between the true background cosmology and the fiducial cosmology used to convert observed angles and redshifts to comoving coordinates. This anisotropy signal is used in BAO analyses to constrain the ratio of the BAO size along versus across the line-of-sight. However, as the AP effect appears at all scales, using the full shape of Ly α forest correlation functions can lead to significantly tighter constraints. Recent eBOSS and DESI full-shape Ly α forest analyses have obtained a factor of ~ 2 tighter constraints on AP when compared to BAO analyses of the same data sets [15, 16]. This significant improvement translates to much tighter constraints on the cosmic expansion rate.

In this work, we use the full shape of the Ly α forest auto-correlation and the Ly α -QSO cross-correlation measured from DESI DR2 data to constrain the AP effect. We use a template-based approach that allows us

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to isolate the AP signal while marginalizing over other effects. The data set used here comprises the first three years of DESI observations [18–24]. The DR2 data set represents an increase of about a factor of two in the number of Ly α forests compared to the first data release (DR1, [25, 26]). Furthermore, Ly α quasars are observed up to six times by DESI, which corresponds to significant improvements in the signal-to-noise of a large fraction of spectra in DR2. In addition to the increase in the sample size, we have also developed several improvements in our analysis methodology which are broadly relevant to Ly α forest analyses.

This work is accompanied by a set of supporting papers that describe certain aspects of the analysis in significantly more detail. These include the validation of the method [27] (hereafter H26), a new set of mocks [28], and the origin of our priors for small-scale non-linearities. [29].

We begin by introducing the data and the Ly α forest correlations we use in Section II. Our methodology is then introduced in Section III and is followed by our validation tests using mocks and blinded data in Section IV. We then present our main AP results in Section V, along with the corresponding cosmic expansion rate constraints. Finally, we explore the cosmological implications of our measurements in Section VI and conclude in Section VII.

II. DATA

We use the same dataset as described in detail in the DR2 Ly α BAO analysis [14] and provide here a brief summary. The DR2 observations were taken between May 2021 and April 2024 and include over 820,000 Ly α forest spectra and the positions of over 1.2 million quasars used for cross-correlations with the Ly α forest, representing approximately a factor of two increase in sample size relative to DR1. In addition, the typical Ly α forest QSO has 2-3 observations in DR2, as opposed to a single observation in DR1. DESI collects spectra with ten identical spectrographs covering 3600–9800 Å in three channels (blue, red, and near-infrared). The blue channel, with spectral resolution $\mathcal{R} \simeq 2000$ –3000, is most relevant for the Ly α forest. All data are processed through the DESI spectroscopic pipeline [22] and classified with the REDROCK software [30, 31], supplemented by a Mg II afterburner [32] and the QUASARNET convolutional neural network classifier [33, 34].

A. Catalogs

The DR2 quasar catalog contains 1,289,874 quasars with $z > 1.77$ that can contribute to the Ly α quasar cross-correlation and 824,989 with $z > 2.09$ with a Ly α forest contributing to the correlations given our selection criteria, following the same construction procedure as for

DR1 [13]. In short, this catalog contains objects observed as part of the main quasar target selection [32] and the WISE variable quasar secondary targeting sample (DESI target bit WISE.VAR.QS0) that are spectroscopically classified as a quasar by either the REDROCK software, the Mg II afterburner, or QUASARNET. A quasar classification with QUASARNET is defined as at least one emission line detected with a confidence greater than 0.99. The catalog is supplemented with quasars incidentally observed as Bright Galaxy Sample (BGS) [35] or Emission Line Galaxy (ELG) [36] targets. For these objects, we apply stricter selection criteria than for main quasar or WISE quasar targets. In both cases we require a quasar classification from REDROCK: an ELG-targeted object must additionally have at least one line detected by QUASARNET with a confidence greater than 0.6, while a BGS-targeted object must additionally have either at least one line detected by QUASARNET with a confidence greater than 0.6 or a quasar classification from the Mg II afterburner. Including quasars from these additional targeting programs beyond the main quasar selection increases the catalog size by roughly 13%.

For this analysis, we updated the Broad Absorption Line (BAL) quasar catalog and the Damped Ly α Absorbers (DLA) catalog from the versions used in the DR2 BAO analysis. Both are used to mask out part of the Ly α forest spectral region when determining the quasar continuum and for the measurement of the correlation functions.

BAL quasars are identified in approximately 20.5% of quasars in the redshift range $1.77 < z < 3.8$ used for the Ly α analysis. We identify and mask the expected locations of BAL troughs falling within the Ly α forest following the same procedure as in [13]. The BAL catalog used here differs very slightly from the one used in the DR2 BAO analysis due to minor code changes that increased the efficiency of the BAL finding tool without altering its algorithm.

We use three methods to identify DLAs [37], based respectively on a Convolutional Neural Network (CNN, [38]), Gaussian Processes (GP, [39]), and Template Fitting (TMP, [40]). Since the BAO analysis, each of the three methods has been improved and re-run, and the full-shape DLA catalog is constructed from these updated results. It is worth noting that the largest change in the behavior of the individual DLA detections is the improvement in the high SNR completeness of the TMP method (see [40]). Using mock catalogs, we updated several combination choices relative to the BAO catalog. First, we increased the redshift matching tolerance between catalogs from 800 to 3000 km s^{−1}. This slightly reduces the overall purity by 0.9% while increasing the completeness by 1%, with a particular gain for DLAs with $\log(N_{\text{HI}}) > 22$ ¹ where completeness improves

¹ N_{HI} is the neutral hydrogen column density.

by 30% with minimal impact on purity. Second, we lowered the CNN detection confidence threshold from $\text{DLA_CONF} > 0.6$ to > 0.5 , which increased both purity and completeness in the mocks. Finally, the BAO analysis only considered DLAs in sightlines with $\text{SNR} > 2^2$, whereas the full-shape catalog includes DLAs in all sightlines. Mocks show that completeness decreases below $\text{SNR} = 2$ while purity remains roughly constant, so the added DLAs maintain the overall purity of the sample.

The impact of new DLA catalog changes are described in Section VB1 of H26. In particular, we tested analysis variations with the DESI DR2 BAO DLA catalog [14, 40] and also with a few different SNR cuts in the new DLA catalog, and found minimal impact on our measurements (see Figure 9 of H26).

B. Ly α forest measurements

We measure the Ly α forest in two regions of the QSO rest-frame spectrum: region B (920–1020 Å) and region A (1040–1205 Å), following the same procedure as in [13] and the DR2 BAO analysis. After applying masks for bad pixels, cosmic rays, BAL troughs, and DLA cores, and discarding forests narrower than 120 Å, the product of mean Ly α flux transmission $\bar{F}(\lambda)$ times the unabsorbed quasar continuum $C_q(\lambda/(1+z_q))$ is estimated by a maximum-likelihood fit that assumes this product to be a universal function of the QSO rest-frame wavelength corrected by a linear polynomial in $\log \lambda$ to account for quasar luminosity, spectral diversity, and redshift evolution of the mean transmission [13]. The transmitted flux overdensity is then

$$\delta_q(\lambda) = \frac{f_q(\lambda)}{\bar{F}(\lambda) C_q(\lambda/(1+z_q))} - 1. \quad (1)$$

Because this procedure subtracts the mean flux and its slope from each individual forest, it introduces distortions in the measured correlation functions.

To model the impact of these distortions, we follow [41] and compute projection matrices η_{ij} for each forest such that $\sum_j \eta_{ij} \delta_q^m(\lambda_j) = \sum_j \eta_{ij} \delta_q^t(\lambda_j)$, where δ_q^t and δ_q^m are the true and measured Ly α forest flux decrements in the forest of quasar q . These projection matrices are given by:

$$\eta_{ij} = \delta_{ij}^K - \frac{w_j}{\sum_k w_k} - \frac{w_j \kappa_i \kappa_j}{\sum_k w_k \kappa_k^2}, \quad (2)$$

where δ_{ij}^K is the Kronecker delta, $\kappa_k = \log \lambda_k - \overline{\log \lambda_k}$, and the weights w_i are the usual Ly α forest pixel weights described in detail by [42]. The η_{ij} matrices multiply the measured δ_q^m before we compute the Ly α correlation

functions. On the other hand, for the theory side we model correlation functions and not δ_q^t . Therefore, the η_{ij} matrices are used to compute distortion matrices, D_{MN} , which multiply the model correlation functions. This is described in more detail in Section III C below.

C. Ly α forest correlations

Following previous DESI Ly α forest analyses, we measure four correlation functions: Ly α (A)×Ly α (A), Ly α (A)×Ly α (B), Ly α (A)×QSO, and Ly α (B)×QSO, where Ly α (A) and Ly α (B) represent Ly α absorption in the A and B regions defined above. We ignore the Ly α (B)×Ly α (B) and QSO×QSO correlations, as well as correlations with higher-order Lyman lines, because they carry less AP information [43]. The correlations are computed on a rectangular grid of comoving separations $(r_{\parallel}, r_{\perp})$ with $4 h^{-1}$ Mpc bins extending to $200 h^{-1}$ Mpc, using the fiducial flat- Λ CDM cosmology from the DR2 BAO analysis [14]. The covariance matrix is estimated from more than 1000 HEALPix sub-regions of the sky, following [13]. To obtain a positive definite covariance matrix, we apply the smoothing procedure described in [13], which has been validated using mocks in previous DESI Ly α forest analyses [e.g., 16, 44, 45]. Some of these tests are replicated in the context of our DR2 full-shape measurements in H26. The effective redshift of the combined sample is $z_{\text{eff}} = 2.33$, in line with previous DESI DR1 and DR2 Ly α forest measurements [13, 14].

The binning results in 50×50 grids for the auto-correlations, and 100×50 grids for the cross-correlations. The larger number of bins in the cross-correlations is due to the use of negative and positive line-of-sight separations, which distinguish between Ly α pixels in front and behind the quasar, respectively. Therefore, our measured data vector contains a total of 15,000 points. However, we impose an isotropic maximum scale cut of $r_{\text{max}} = 200 h^{-1}$ Mpc, which reduces the number of fitted points to 11790. Minimum-scale cuts are applied using a new small-scale marginalization method, which will be described in Section III C. The reasoning and tests that led to our scale cuts are described in Section IV A below.

III. METHODOLOGY

In this section, we describe our methodology for measuring the Alcock–Paczynski effect from Ly α forest correlation functions. Our approach closely follows that used for the same measurement in DESI DR1 [16] but incorporates model improvements developed for the DR2 BAO analysis [14] and also two new improvements which were developed during this project.

We begin with a detailed description of the Alcock–Paczynski effect in Section III A. After that, we give an overview of our model for the Ly α forest correlation functions and its contaminants in Section III B. Finally, we

² Measured over the rest-frame wavelength range $1420 < \lambda_{\text{rest}} < 1480$ Å.

present the two new methodology improvements in [Sections III C](#) and [III D](#).

A. The Alcock–Paczyński effect

1. Modeling and parametrization

The AP effect is the apparent stretching or compression of intrinsically isotropic cosmic structures that occurs when angles and redshifts are converted into distances using an incorrect cosmological model. For a 3D tracer, its correlation function in comoving coordinates, $\xi(r_{\parallel}, r_{\perp})$, is derived from measurements of angular separations $\Delta\theta$ and redshift pairs (z_i, z_j) , with:

$$r_{\parallel} = [D_C(z_i) - D_C(z_j)] \cos \frac{\Delta\theta}{2}, \quad (3)$$

$$r_{\perp} = [D_M(z_i) + D_M(z_j)] \sin \frac{\Delta\theta}{2}, \quad (4)$$

where the comoving distance, $D_C(z)$, and the comoving angular diameter distance, $D_M(z)$, are computed based on a fiducial background cosmology (see [Table I](#)). When fitting a model, we can account for this effect by introducing two scale parameters, $q_{\parallel}$ and $q_{\perp}$, which rescale the coordinates along and across the line-of-sight, respectively. These parameters account for the ratio between the coordinates obtained using the fiducial cosmology ($r_{\parallel}, r_{\perp}$) and the coordinates we would have obtained using the true cosmology ($r'_{\parallel}, r'_{\perp}$), such that $r'_{\parallel} = q_{\parallel} r_{\parallel}$ and $r'_{\perp} = q_{\perp} r_{\perp}$. In this work, we follow [\[43\]](#) and isolate the AP anisotropy using the parameter:

$$\phi \equiv \frac{q_{\perp}}{q_{\parallel}} = \frac{D_M/D_H}{[D_M/D_H]_{\text{fid}}}, \quad (5)$$

where $D_H = c/H$, with the Hubble parameter H , and speed of light c . Therefore, ϕ measures the ratio of the true AP parameter $F_{\text{AP}} = D_M/D_H$ with respect to its fiducial value. Note that this measurement does not depend on a specific scale, such as BAO, since the AP effect distorts all scales in our measured correlations equally. For more details on this parametrization and the motivation behind this approach, see [\[43\]](#).

In BAO analyses [e.g., [14](#)], the acoustic scale is constrained using a template approach based on an isotropic matter power spectrum $P_{\text{fid}}(k)$. This template is decomposed into a peak (or wiggle) component and a smooth (or no-wiggle) component. The AP effect from the BAO scale is then measured by rescaling the peak component anisotropically, while the coordinates of the smooth components are kept fixed. Here, we extend this framework by also rescaling the coordinates of the smooth component, thus extracting the AP effect from the full shape of $\text{Ly}\alpha$ forest correlations. This approach was first introduced in the context of $\text{Ly}\alpha$ forest analyses by [\[43\]](#), and we follow their methodology here. Finally, we note

Parameter	Planck 2018 cosmology (TT,TE,EE+lowE+lensing)
$\Omega_m h^2 =$	0.14297
$+\Omega_c h^2$	0.12
$+\Omega_b h^2$	0.02237
$+\Omega_{\nu} h^2$	0.0006
h	0.6736
n_s	0.9649
$10^9 A_s$	2.100
Ω_m	0.31509
Ω_r	7.9638e-05
$\sigma_8(z=0)$	0.8119
$f(z_{\text{eff}}=2.33)$	0.9703
$f\sigma_8(z=2.33)$	0.2979
r_d [Mpc]	147.09
r_d [h^{-1} Mpc]	99.08
$D_H(z_{\text{eff}}=2.33)/r_d$	8.6172
$D_M(z_{\text{eff}}=2.33)/r_d$	39.1879

TABLE I. The fiducial cosmology parameters used in this work. The first set of parameters defines the cosmological model in Λ CDM, while the second set contains derived quantities used in this work. The fiducial cosmology is used to convert observed angles and redshifts into comoving coordinates, and to compute the template matter power spectrum at the core of our model for $\text{Ly}\alpha$ correlations.

that we use the terms smooth component and broadband interchangeably in this work.

The parametrization we use for our scale parameters is given by the ϕ parameter above and by an isotropic scale parameter α , defined as $\alpha \equiv \sqrt{q_{\perp} q_{\parallel}}$. We choose this particular combination because it is close to the optimal combination that minimizes the uncertainty in the isotropic BAO signal from the $\text{Ly}\alpha$ forest [see e.g., [13](#), [43](#)]. This choice does not impact our results in any way, and is purely a matter of preference.

Because our template still uses a peak and a smooth component (as in BAO analyses), we introduce two sets of scale parameters: one for the peak component (ϕ_p, α_p) and one for the smooth component (ϕ_s, α_s). As described above, the interpretation of the AP parameters is consistent for both components. In the case of α_p , the measurement corresponds to:

$$\alpha_p \equiv \sqrt{q_{\perp} q_{\parallel}} = \frac{\sqrt{D_M D_H}/r_d}{[\sqrt{D_M D_H}/r_d]_{\text{fid}}}, \quad (6)$$

where r_d is the size of the acoustic scale at the drag epoch. In principle, the isotropic scale from the smooth component, α_s , also contains cosmological information, likely related to the scale of matter-radiation equality [\[43\]](#). However, this measurement is fairly weak and affected

by many of the contaminants in the Ly α forest. Therefore, we treat α_s as a nuisance parameter and marginalize over it, in line with previous analyses [15, 16]. As the parameters for the BAO peak (ϕ_p, α_p) have already been measured in the DR2 BAO analysis [14], our main goal here is to also measure the AP effect from the broadband through the ϕ_s parameter. Our final results will include consistent measurements of all three parameters: ϕ_s, ϕ_p, α_p .

2. Alcock–Paczynski versus RSD

In practice, the AP effect is just one of multiple effects that introduce anisotropies in our measurements. Therefore, in order to measure the AP effect, we need to marginalize over other sources of anisotropy. The primary anisotropy present in our correlations is due to redshift-space distortions (RSD). When the anisotropy due to RSD has a functional form similar to that introduced by the AP effect, marginalizing over RSD results in a significant loss of AP information. Indeed, when the two functional forms are identical, the effects are completely degenerate, and we cannot extract information from either.³ On the other hand, when there is a strong scale dependence in our measured correlation functions or power spectra, the two effects are fairly distinct. This is because scale-dependent features like BAO allow us to distinguish between a coordinate change (AP) and an amplitude change (RSD), even when the two anisotropies have the same functional form.

We illustrate the similarities and differences between RSD and the AP effect in Figure 1 by plotting the response of a linear theory (Kaiser) auto-correlation model in configuration space to changes in the RSD amplitude parameter β (left column) and to small changes in the broadband AP parameter ϕ_s (right column). For this comparison, we use only the smooth component (i.e., without BAO) and show each response for models with three different values of β , including $\beta = 0$ (no RSD), $\beta = 0.4$ (typical value for galaxy clustering), and $\beta = 1.2$ (typical value for the Ly α forest). We also mask the small scales ($r < 30 h^{-1}$ Mpc), as in that regime linear theory breaks down, and the comparison would need to include marginalization over non-linear effects.⁴

In the case of null RSD, the responses to the two parameters have very similar shapes to each other, indicating that the two effects are close to degenerate. In

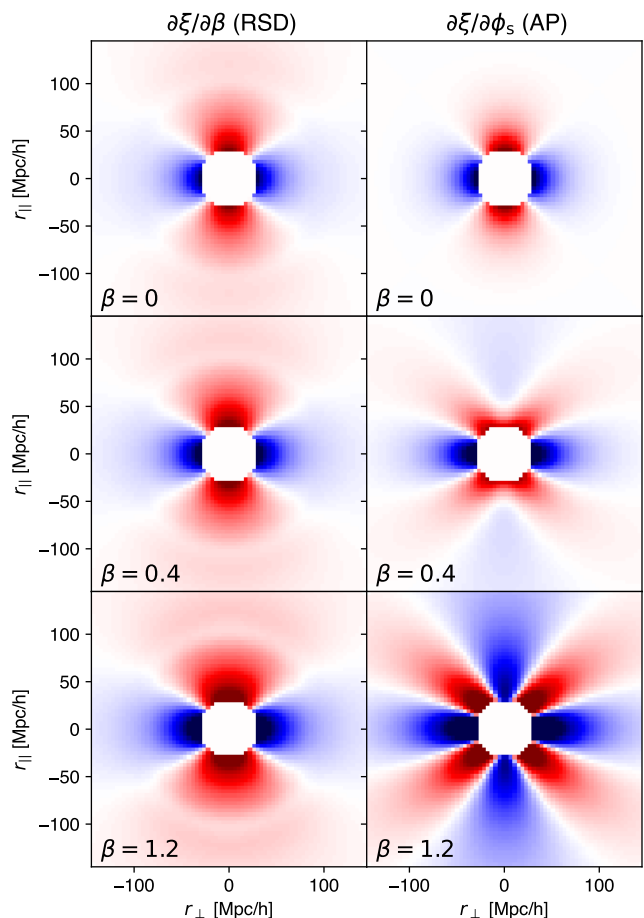


FIG. 1. Illustration of parameter sensitivity in configuration space for a linear theory model, focusing on the smooth component only (i.e., no BAO). Each row shows a different fixed value of the RSD parameter β , while the left and right columns show the response to small changes in β and in the broadband AP parameter ϕ_s , plotted as functions of transverse and line-of-sight separations. The response to β is dominated by LOS-versus-transverse anisotropy and preserves its shape with increasing RSD. On the other hand, the response to ϕ_s evolves from a shape similar to the one for β at null RSD to a significantly different shape as β increases. Thus, the RSD and AP signals become easier to disentangle for tracers with larger values of β .

fact, one can show that in the case of a purely scale-independent (i.e., power law) power spectrum, the two effects are exactly degenerate in the limit of null RSD.⁵ On the other hand, as β increases, the response to AP evolves quite significantly, while the response to RSD preserves its shape. Furthermore, with increasing β , the amplitude of the response to AP also significantly increases, which

³ Unless an external constraint on one of the quantities is used to break the degeneracy.

⁴ The exact scale where this happens can be quite different depending on the tracer and redshift used. We only use this as an example to illustrate why AP and RSD are less degenerate for Ly α , so we refrain from a more realistic investigation in the context of galaxy tracers, and use the $r > 30 h^{-1}$ Mpc scale cut (which we have validated for the Ly α forest at redshifts $2 < z < 4$) for all comparisons.

⁵ While we use the smooth component here, this still has some scale dependency at low- k due to the turn-over in the matter power, which explains why the two responses are not exactly identical here.

is illustrated through darker colors in the bottom right panel versus the top right panel of [Figure 1](#) (the same color scale is used for each column). For galaxy clustering, the relevant comparison is the one in the middle row, where the two responses still have a fairly similar pattern at small separation where amplitudes of the derivatives are largest. Finally, the comparison relevant for the Ly α forest is the one in the bottom row, where the response to AP has a very distinct pattern from the one of the response to RSD.

The comparisons in [Figure 1](#) show that AP is easier to disentangle from RSD in Ly α forest analyses compared to galaxy clustering analyses. This is one of the key reasons why Ly α forest analyses have so far delivered significant improvements (a factor of 2 to 2.4 better) in the AP constraints from full-shape analyses versus BAO analyses of the same datasets [[15](#), [16](#)]. On the other hand, galaxy clustering analyses have only delivered mild improvements (10% to 30% better) in full-shape AP compared to BAO [e.g., [46](#), [47](#)], although it is important to note that modeling non-linearities also plays a more important role in that case.⁶

B. Modeling Ly α forest correlations

Our model of the Ly α forest correlation functions closely follows that used in [[14](#)] and [[16](#)], and is implemented in the *Vega* package⁷ [[48](#)]. The anisotropic Ly α auto- and Ly α -QSO cross-power spectra are given by:

$$P_F(\mathbf{k}) = b_F'^2(k) (1 + \beta_F'(k) \mu_k^2)^2 \times G(\mathbf{k}) F_{NL}(\mathbf{k}) P_{\text{fid}}(k), \quad (7)$$

$$P_{F \times Q}(\mathbf{k}) = b_F'(k) (1 + \beta_F'(k) \mu_k^2) (b_Q + f \mu_k^2) \times G(\mathbf{k}) X_{NL}(\mathbf{k}) P_{\text{fid}}(k), \quad (8)$$

where $\mathbf{k} = (k_{\parallel}, k_{\perp})$, $\mu_k = k_{\parallel}/k$, and $P_{\text{fid}}(k)$ is computed with CAMB [[49](#)] at $z_{\text{eff}} = 2.33$ using the fiducial *Planck* cosmology in [Table I](#). The function $G(\mathbf{k})$ accounts for the finite bin size ($4 h^{-1}$ Mpc) of the correlation function grids and is given by a sinc function [[12](#)]. The effective Ly α bias b_F' and RSD parameter β_F' account for both Ly α flux and undetected HCD absorption and are a function of $k_{\parallel}$ (see [Section III B 3](#)). They also have an extra scale-dependent contribution due to UV background fluctuations (see [Section III D](#)). b_Q is the linear quasar bias and f is the logarithmic derivative of the linear growth factor. F_{NL} and X_{NL} model small-scale non-linearities in the auto- and cross-correlations respectively, and are described below. These anisotropic power

spectra are transformed into 2D model correlation functions via a multipole decomposition up to $\ell = 6$ followed by a Hankel transform.⁸

Note that in [Equations \(7\) and \(8\)](#), we made explicit the dependency of the QSO term on f , but used the RSD parameter β_F for the forest terms. This is because the Ly α forest RSD parameter is given by $\beta_F = b_{\eta} f / b_F$, which includes an extra unknown velocity divergence bias, b_{η} . The presence of b_{η} precludes direct measurements of the cosmic growth rate from linear scales in the Ly α forest, unless other sources of information are used to constrain it and break its degeneracy with f (e.g., from hydro-simulations [[50](#)]). Therefore, we follow previous Ly α forest analyses and directly marginalize over β_F [[15](#), [16](#)]. We also note that in early versions of our analysis, we attempted to measure the growth rate times the amplitude of fluctuations in $8 h^{-1}$ Mpc regions, $f \sigma_8$, from the QSO term in the cross correlation, following [[16](#), [43](#)]. However, as in DESI DR1 [[16](#)], we decided to de-scope that measurement before unblinding because we found it was not robust in either mocks or data. Given that the constraint is fairly weak ($\sim 15\%$), we chose to simply treat the growth rate as a nuisance parameter to be marginalized. See Appendix E of [H26](#) for more details.

1. Small-scale non-linearities

For the Ly α auto-correlation, we model small-scale deviations from linear theory with the empirical model of [[51](#)]:

$$F_{NL}(\mathbf{k}) = \exp \left\{ [q_1 \Delta^2(k) + q_2 \Delta^4(k)] \times \left[1 - \left(\frac{k}{k_v} \right)^{a_v} \mu^{b_v} \right] - \left(\frac{k}{k_p} \right)^2 \right\}, \quad (9)$$

where $\Delta^2(k) \equiv k^3 P_{\text{fid}}(k) / (2\pi^2)$ and $\{q_1, q_2, k_v, a_v, b_v, k_p\}$ are either treated as free parameters or fixed to values measured from other datasets. This model was shown to accurately reproduce small-scale Ly α power spectra in hydrodynamical simulations [[51](#), [52](#)].

Previous Ly α forest full-shape analyses [[16](#)] marginalized over various combinations of parameters in this model. However, we found that after introducing marginalization over small scales in the undistorted correlations, as described in [Section III C](#) below, we lost sensitivity to these parameters.⁹ Therefore, we decided to fix them in our baseline model to the values measured in the companion paper [[29](#)]: $q_1 = 0.303$, $q_2 = 0.267$,

⁶ Both due to galaxies tracing highly clustered environments when compared to the Ly α forest, and because large galaxy surveys have so far been done at lower redshifts.

⁷ <https://github.com/andreicuceu/vega>

⁸ We have tested that going up to $\ell = 10$ does not have a significant impact on our results.

⁹ By loss of sensitivity we mean that the posterior does not improve upon the prior for these parameters. See [H26](#) for more details.

$k_v = 0.576$, $a_v = 0.443$, $b_v = 1.66$, $k_p = 11.062$. These constraints come from measurements of the Ly α forest 1D power spectrum [53–55] and used the ForestFlow emulator trained on hydrodynamic simulations [50].¹⁰ We tested the impact of this choice and found that it does not significantly affect either the values nor the uncertainties of our measurements. This is discussed in more detail in Section IV B, and in H26.

2. Quasar redshift errors

For the cross-correlation, the dominant small-scale effect is due to quasar redshift errors [56], which we model as a Lorentzian damping of the cross-power spectrum:

$$X_{NL}^2 = [1 + (k_{||}\sigma_z)^2]^{-1}, \quad (10)$$

where σ_z is a free parameter. In practice, this model is also sensitive to the impact of large peculiar velocities (finger-of-god effect). H26 tested using a Gaussian damping instead of the Lorentzian, and found no significant impact on our constraints. The fact that our results remain insensitive to the exact functional form of this damping is a good indication that our scale cuts in the cross-correlation, $r > 40 h^{-1}$ Mpc, are conservative enough.

In addition to the damping effect, we also account for the impact of systematic redshift errors. The Ly α -QSO cross-correlation has both negative and positive $r_{||}$ values to distinguish between Ly α absorption in front versus behind the quasar it is correlated with (Section II). Systematic quasar redshift errors lead to a shift of the measured cross-correlation along the $r_{||}$ direction. We account for this possible shift with an extra parameter, $\Delta r_{||}$, that shifts the coordinate grid of the cross-correlation along the line-of-sight.

3. High column density (HCD) absorbers

Strong HCDs ($\log N_{\text{HI}} > 20.3$, i.e. DLAs) are masked as described in Section II. Weaker HCDs, and those in low-SNR spectra which have low DLA-finder completeness [40], contribute to the correlations through effective Ly α bias and RSD parameters [12, 57]:

$$b'_F = b_F + b_{\text{HCD}} F_{\text{HCD}}(k_{||}), \quad (11)$$

$$b'_F \beta'_F = b_F \beta_F + b_{\text{HCD}} \beta_{\text{HCD}} F_{\text{HCD}}(k_{||}), \quad (12)$$

where b_{HCD} and β_{HCD} are the HCD linear bias and RSD parameter. As in the BAO measurement [14], we adopt the exponential profile $F_{\text{HCD}} = \exp(-L_{\text{HCD}} k_{||})$, where L_{HCD} is a free parameter representing the typical scale of undetected HCDs.

4. Metal contamination

Metal absorption near the Ly α rest-frame wavelength in the Ly α forest produces spurious peaks along the line of sight in the correlation functions because we treat all forest pixels as Ly α pixels when computing their redshift [58, 59]. We model the contribution of four silicon lines (Si II(1207), Si II(1190), Si II(1193) and Si II(1260)), modeling their cross-correlation with Ly α and the other metal lines, along with their auto-correlation. We also include the CIV auto-correlation from smaller redshifts. Each metal line is treated as an independent tracer of LSS and has its own free linear bias parameter and an RSD parameter fixed to 0.5, following [41, 60].¹¹ For each contributing pair of transitions, the contaminated model correlation is mapped from the true comoving separation to the observed grid via a *metal* matrix (see Eq. 10 and 11 in [16]).

5. Other effects

We also include two additional contamination terms. First, spectra from fibers of the same spectrograph share correlated sky noise, which contributes primarily at purely transverse separation ($r_{||} = 0$). We model this with a fixed template $\xi_{\text{noise}}(r_{\perp})$ and a free amplitude a_{noise} , following [13, 60].

Second, for the Ly α -QSO cross-correlation we include the transverse proximity effect [56], modeled as an isotropic profile

$$\xi^{\text{TP}} = \xi_0^{\text{TP}} ((1 h^{-1} \text{Mpc})/r)^2 \exp(-r/\lambda_{\text{UV}})$$

with a free amplitude ξ_0^{TP} and fixed scale $\lambda_{\text{UV}} = 300 h^{-1}$ Mpc.

C. Small-scale marginalization

As mentioned in Section IIB, the measured Ly α forest correlations are distorted by the fact that we fit for an amplitude and slope in the continuum of each forest, which has the effect of removing LSS modes of the size of the forest and larger [61]. To model the impact of this distortion on our model correlations, we compute distortion

¹⁰ We note that pixel pairs in the same forest are not included in our 3D correlation measurements, which means there is no overlap between the data used here and that used by [29] to obtain these constraints. Furthermore, this information comes from scales that are significantly smaller than those used in our analysis [55].

¹¹ We have tested that fixing the metal RSD parameters to 0.4 and 0.6 does not lead to significant changes in our constraints.

matrices, D_{MN} , which multiply the correlation function model, $\xi(r_{\parallel}, r_{\perp})$, to give the distorted correlation:

$$\hat{\xi}_M = \sum_N D_{MN} \xi_N, \quad (13)$$

where N indexes over undistorted model bins, and M over data bins. The distortion matrices for the Ly α auto and QSO cross-correlation are given by:

$$D_{MN}^{\text{auto}} = \frac{1}{W_M} \sum_{i,j \in M} w_i w_j \sum_{i',j' \in N} \eta_{ii'} \eta_{jj'} f_a(z_i, z_j, z_{\text{ref}}), \quad (14)$$

$$D_{MN}^{\text{cross}} = \frac{1}{W_M} \sum_{i,j \in M} w_i w_j \sum_{i',j' \in N} \eta_{ii'} f_c(z_i, z_j, z_{\text{ref}}), \quad (15)$$

where $W_M = \sum_{i \in M} w_i$, and the model bins N are half the size (i.e., $2 h^{-1}$ Mpc) of our data bins M . The functions f_a and f_c account for the evolution of the amplitude of the correlation functions between a reference redshift z_{ref} and the redshifts of each measurement [62].

The impact of multiplying the model correlations with the distortion matrix has been studied in detail in [41, 63]. The primary effect is a mixing of information along the line-of-sight in the correlation function (i.e., as a function of $r_{\parallel}$). During our validation process, we encountered a consequence of this mixing which, if ignored, could lead to percent-level biases in Ly α AP constraints (see Section A). In particular, we found that this mixing results in significant information from very small $r_{\parallel}$ scales in ξ_N (i.e. $r_{\parallel,N} < 10 h^{-1}$ Mpc) leaking into all scales of $\hat{\xi}_M$ (i.e., including $r_{\parallel,M} > r_{\min}$). This means that our usual scale cuts $r > r_{\min}$ do not have the intended effect of completely removing our sensitivity to scales $r < r_{\min}$. To resolve this issue, we introduce a new formalism for marginalizing over this small-scale contamination.

The first step in this new approach is to marginalize over small-scale bins at $r < r_{\min}$ in the undistorted model space, ξ_N . We do this by adding one free amplitude parameter η_T for each bin T in the undistorted model, where the bins T do not have to be the same as the native undistorted model bins N . Our choice of bins T is such that they match the data bins M that are coarser than the model bins. The amplitudes η_T multiply a linear template in the distorted space, given by $\hat{\Delta}_{MT} = \sum_N D_{MN} \Delta_{NT}$, where $\Delta_{NT} = 1$ if the finer bin N is included in the coarser data bin T and 0 elsewhere. In the case of linear templates, the most efficient method to find η_T is to analytically solve for the template amplitudes that maximize the posterior density (i.e., minimize the χ^2). We denote all model parameters with θ and the chi-squared with a general Gaussian prior is:

$$\chi^2 = (\xi_{\text{obs}} - \hat{\xi}_{\text{model}}(\theta))^T C^{-1} (\xi_{\text{obs}} - \hat{\xi}_{\text{model}}(\theta)) + (\theta - \mu)^T S^{-1} (\theta - \mu), \quad (16)$$

with a mean μ and covariance matrix S , where $S_{ii} \rightarrow \infty$ is equal to an uninformative prior on a given parameter

θ_i . Omitting the template indices T , we denote the vector of linear template amplitudes η , and the associated templates with a matrix $\hat{\Delta}$, such that $\hat{\Delta}\eta$ results in a vector of the same shape as the data vector ξ_{obs} . Additionally, the model parameters θ can absorb the mean μ , which is set to zero for simplicity. With this definition, η parametrizes deviations from the model within each bin. A modified covariance matrix $\tilde{C} = C + \hat{\Delta} S \hat{\Delta}^T$ captures the marginalized chi-squared:

$$\chi_{\text{marg}}^2 = (\xi_{\text{obs}} - \hat{\xi}_{\text{model}}(\theta))^T \tilde{C}^{-1} (\xi_{\text{obs}} - \hat{\xi}_{\text{model}}(\theta)). \quad (17)$$

Note that this formulation front-loads the marginalization into a covariance matrix update at the beginning of the analysis, so that the best-fitting η_{MAP} are not computed per model evaluation. These are instead computed at the end as follows:

$$\eta_{\text{MAP}} = (\hat{\Delta}^T C^{-1} \hat{\Delta} + S^{-1})^{-1} (\hat{\Delta}^T C^{-1} r), \quad (18)$$

where $r \equiv \xi_{\text{obs}} - \hat{\xi}_{\text{model}}(\theta_*)$ and θ_* minimizes χ_{marg}^2 .

In practice, we add as many templates (rows of $\hat{\Delta}$) and their associated amplitude parameters (η) as there are data bins at $r < r_{\min}$. Though this introduces many more parameters, we also retain all data at $r < r_{\min}$ in the χ^2 fit such that each template amplitude parameter is well constrained by the corresponding data bin. Consequently, this marginalization technique does not add degrees of freedom when compared to a fit without small-scale marginalization and the inclusion of data at $r \geq r_{\min}$ only. Rather, it ensures that we correctly model the leakage of small-scale correlations at larger scales caused by the spectral continuum fitting and encoded in the distortion matrix.

D. UV background fluctuations

Past Ly α forest analyses were based on the assumption of a homogeneous ionizing UV background (UVB). However, work by [64] and [65] has shown that fluctuations in this ionizing background can give rise to a scale-dependent Ly α bias, where the relevant scales are given by the mean-free path of UV ionizing radiation, which is expected to be of order hundreds of Mpc at redshift 2.3. During our validation process, we tested including a model for this effect (as was done in previous analyses, e.g. [16, 62]) and found that it is detected at high statistical significance ($\sim 3\sigma$ to $\sim 5\sigma$ depending on the exact model setup). Therefore, we decided to include it in our baseline model. We use the simplest version of the model, which only includes the scale-dependent Ly α bias, as described below.

We model the impact of UVB fluctuations following [65], which started by introducing fluctuations in the photoionization rate $\Gamma(x)$, defined as $\delta\Gamma(x) = \Gamma(x)/\bar{\Gamma} - 1$, and showed that they give rise to a scale-dependent Ly α bias,

given by:

$$b_F(k) = b_\delta + b_\Gamma \frac{(b_s - b_a)W(k/\kappa_0)}{1 + b'_a W(k/\kappa_0)}, \quad (19)$$

where $W(s) = \arctan(s)/s$, b_s is the bias of the UV sources, b_a is the bias of UV absorbers, b'_a is the response of these absorbers to changes in the ionizing intensity, and $\lambda_0 = 1/\kappa_0$ is the mean free path of ionizing photons.

We treat b_Γ as a free parameter and fit it. We follow [65] and adopt $(b_s - b_a) = 1$, as this directly multiplies b_Γ , and hence we are only sensitive to the product $b_\Gamma(b_s - b_a)$. We also adopt the values $b'_a = -2/3$ and $\lambda_0 = 300h^{-1}$ Mpc, following [65]. We have tested small changes around these values, and found they do not have a significant impact on our fit.

The prediction of [65] is that $b_\Gamma \simeq 0.13$ at $z = 2.25$. We measure $b_\Gamma = 0.143 \pm 0.047$ at $z_{\text{eff}} = 2.33$, which is in excellent agreement with the prediction. We have also tested freeing λ_0 , and found results in agreement with $\lambda_0 \sim 300h^{-1}$ Mpc, but with very large uncertainties. While the presence of UVB fluctuations in our model significantly improves the fit, they do not have a significant impact on our AP constraints, as discussed in Section IV B, and shown in H26.

Finally, we note that while only the simplest version of this model is included in our baseline analysis, we have also tested including additional terms and found that the model does not significantly improve with them. These include the shot noise of the UVB sources [65], and the effect of the HeII re-ionization, which is expected to produce a much weaker scale dependence in the Ly α bias on scales of $\sim 30h^{-1}$ Mpc [65]. These extensions will be explored in more detail in future work.

IV. ANALYSIS VALIDATION

Our analysis validation process has two major components: validation of the baseline analysis and model using synthetic data sets (mocks), and robustness tests on blinded data. Both of these components are described in detail in H26. Here, we provide an overview of the validation process and highlight the main results, which demonstrate that our measurements are robust. We present the validation with mocks in Section IV A, the blinded robustness tests in Section IV B, and discuss our results and conclusions in Section IV C.

We primarily focus our validation on what is new in this work relative to DR2 Ly α BAO [14], namely the broadband AP constraint, ϕ_s . This is because our analysis approach involves a template with a peak and a smooth component whose coordinates are independently re-scaled (see Section III A), which makes it almost identical to a BAO analysis when it comes to constraining the position of the BAO peak.¹² Our analysis builds

upon the DESI DR2 BAO analysis [14] by also rescaling the coordinates of the smooth component, although we have checked the BAO results for all of our tests on both mocks and blinded data to ensure there are no significant changes with respect to the results presented in [14, 45]. Appendix C of H26 presents the BAO results from our blinded data tests, while the BAO tests on mocks are briefly discussed in Section IV A below, and in more detail in Section IV of H26.

The DR2 full-shape analysis was blinded in the same manner as the DR1 full-shape analysis described in [16]. The process consisted of adding a random value to the parameter of interest, namely ϕ_s , within the likelihood. The random offset was generated from a Gaussian with zero mean and standard deviation of 2%, based on the eBOSS measurement of [15]. The random offset was implemented in our modeling package and automatically kept constant across all of the tests we performed.

The overarching goal of our validation process was to identify a set of analysis configurations that allow us to measure the AP effect primarily at large scales where the linear-theory approach works well. This means that any systematic uncertainties from small-scale non-linearities or contaminants need to be kept at a level well below our statistical uncertainties. If we found this is not possible, we would increase our final uncertainties to account for the systematic error budget. Following previous DESI Ly α forest analyses [13, 14, 16], our threshold for validating the measurements was set at 1/3 of the statistical uncertainty. For ϕ_s , this corresponds to a tolerance of 0.38%. For the mock validation, we required the baseline results to be within this threshold and also to display convergence with more conservative choices.¹³ For the blinded analysis and modeling variations, we required results to be within this threshold. Any variations exceeding this threshold had to be investigated in detail and addressed before unblinding our measurements. Finally, we also ran a set of data splits to ensure consistency across different subsets of our data, and for these we required results to be within 2σ of each other.

All decisions on what constitutes our baseline analysis were made using blinded data. Once we converged on this baseline analysis, ensured mock results were within the set tolerance, and investigated and addressed any variations outside the tolerance and data splits in tension, we prepared a detailed technical document describing our tests and validation process.¹⁴ This document was then circulated to the entire DESI collaboration, which had a one-week period to provide comments and feedback before we proceeded to unblind our measurements. We have

for Ly α forest BAO analyses.

¹³ This is to ensure that we do not cherry-pick an analysis configuration where systematic errors happen to cancel each other out.

¹⁴ Most of the details of this document are presented in the companion paper H26.

¹² This is also because we do not currently use reconstruction [66]

made no changes to the baseline analysis post-unblinding, and the results presented here are identical to those from when we unblinded.

A. Validation with Mocks

Our validation with mocks was based on two different types of mocks. The first set of Ly α forest mocks were developed from the **AbacusSummit** simulations by [67] to test the sensitivity of our model for the Ly α auto-correlation and Ly α QSO cross-correlation functions to non-linearities in the clustering at small scales. These mocks only simulate the Ly α forest and quasar clustering and use skewers drawn from a cubic box at redshift $z = 2.5$. The second set of mocks are **CoLoRe 2LPT** mocks, which were produced with the **CoLoRe** [68] package using Second Order Lagrangian Perturbation Theory (2LPT). These mocks are described in detail in a companion paper [28], and we only give a brief overview below. Both sets of mocks use the fluctuating Gunn-Peterson approximation [FGPA; 69] to estimate the Ly α forest from the matter density field.

The **AbacusSummit** simulations are a suite of large, high-accuracy N-body simulations designed to meet the cosmological simulation requirements of the DESI survey [70, 71]. The Ly α signal in these mocks was computed by [67] according to four models, which we label M1 to M4 (see Table 1 of [67]). These models correspond to different parameters and methods to estimate the Ly α optical depth from the smooth matter density field in the **ABACUSSUMMIT** simulations. While the simulations are not as realistic as hydrodynamical simulations, they include the effect of non-linear clustering inherited from the matter field in the original N-body simulation, and the variations in the optical depth modeling give us some sense of the importance of those assumptions for the signal we want to measure. We primarily use these mocks to test our baseline model for the Ly α auto- and cross-correlation with quasars and to validate our scale cuts. These mocks are also useful for testing biases in the BAO measurements due to non-linear evolution [72, 73].

The **CoLoRe-2LPT** mocks are light-cone simulations that replace the standard **CoLoRe** log-normal prescription [68] used in previous Ly α forest analyses with a 2LPT approximation for both the density and velocity fields. This new prescription captures mild non-linearities and the broadening of the BAO peak, which were absent from previous mocks. The QSO clustering model is also improved, as it has been calibrated with HOD catalogs from the **Abacus** N-body simulations and includes the Fingers-of-God effect. Line-of-sight 2LPT density and velocity skewers are drawn from these full-sky light-cones and converted into Ly α transmission skewers using a new 2LPT version of **LyaCoLoRe** [74] that has been calibrated to reproduce the Ly α clustering statistics of the latest DESI measurements. We then produce 400 DESI DR2 mocks by post-processing these skewers with the

quickquasars script from the **desisim** package, which applies the DESI survey footprint, generates quasar continua, adds astrophysical contaminants (metals, BALs, and DLAs), and simulates instrumental noise and the spectrograph resolution. Finally, we analyse the mocks with the same pipeline used for the real data. The new mocks are presented in detail in the companion paper [28].

Our process for validating the analysis using mocks consisted of two steps. First, we used the **AbacusSummit** mocks and the stack of correlations from the 400 **CoLoRe 2LPT** mocks to identify an analysis configuration that passes our validation criteria or minimizes the combined statistical and systematic uncertainty. The primary choices during this step were to select a set of scale cuts (discussed below), a configuration for small-scale marginalization (Section III C), and the model and priors for contaminants (Sections B and III B). This first step was conducted in parallel with a wide range of tests on blinded data to ensure that the choices we made also yield robust measurements on real data. In particular, an important goal of this first step was to identify a model that provides good fits to both mocks and real data, while also providing unbiased constraints.

In the second step, we used the baseline model found during the first step to fit all individual **CoLoRe 2LPT** mocks and then performed statistical tests on the mock population. The main goal of the second step was to validate our uncertainties. In the remainder of this section, we provide an overview and the main results of the first step in this process, and refer the reader to the companion paper H26 for details on the rest of the tests we performed.

As described in Section II, with the exception of the updates to the DLA catalog, the analysis choices up to and including the measurement of the correlation functions are the same here as in the previous DESI DR2 Ly α forest analysis [14]. This is because we managed to find a set of model configurations that yielded unbiased fits in mocks (up to our threshold), so we did not have to reconsider changes to analysis choices at an earlier stage.

The most important choice was the minimum scale included in the data we fit, $r_{\min}$.¹⁵ In Figure 2, we show mock results on the broadband AP parameter, ϕ_s , relative to the true value of ϕ_s in mocks, for different values of $r_{\min}$. These results are obtained from joint fits of the Ly α auto-correlation and its cross-correlation with quasars. The gray bands indicate the 1/3 threshold of the statistical uncertainty in the data. For this test, we also ran

¹⁵ One could also consider minimum scale cuts in the line-of-sight or transverse separations, $r_{\parallel}$ and $r_{\perp}$. We did perform a few tests with scale cuts in $r_{\parallel}$, $r_{\perp}$ as part of our validation (see H26). However, these types of cuts are undesirable in AP analyses, as the goal is to measure the anisotropy in the correlation function, and any significant cuts in $r_{\parallel}$, $r_{\perp}$ would remove a large part of the signal.

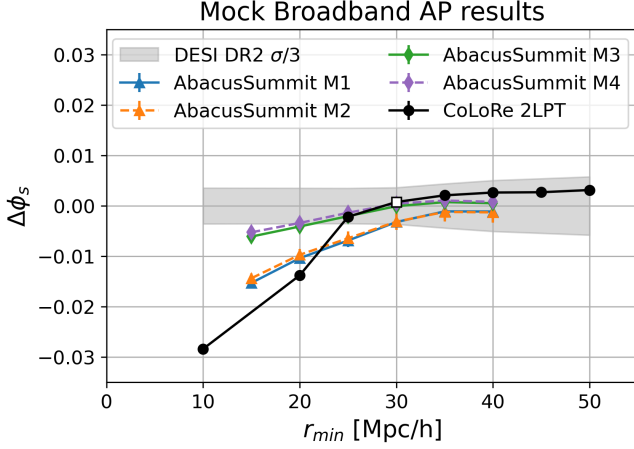


FIG. 2. Broadband AP constraints in mocks relative to the truth as a function of the minimum separation used in the analysis, $r_{\min}$. The gray bands indicate the $1/3 \sigma_{\text{DR2}}$ threshold obtained from the joint analysis of all four DESI DR2 correlations. We show results from fits to the four different versions of **AbacusSummit** mocks, and from the stacked correlations of 400 **CoLoRe 2LPT** mocks. The baseline analysis configuration is shown as an empty square.

the same fits on the real data, so we can compare mock results with the relevant threshold for each $r_{\min}$ value. When including smaller scales around 10 or $20 h^{-1}$ Mpc, we observe significant systematic biases in ϕ_s of up to 3%, and results that rapidly change with different $r_{\min}$. On the other hand, for $r_{\min} \geq 30 h^{-1}$ Mpc, both the **CoLoRe 2LPT** mocks and all four types of **AbacusSummit** mocks show converging results within our thresholds. Therefore, we conclude that our model is unbiased within the thresholds we imposed when using scales larger than $30 h^{-1}$ Mpc.

We show more detailed versions of this test in [H26](#), where we vary $r_{\min}$ independently for the auto and cross-correlations. We have also tested changing the maximum scale cut $r_{\max}$ and found that it does not have a significant impact on AP constraints. Based on these results, we chose $r_{\min} = 30 h^{-1}$ Mpc for the $\text{Ly}\alpha$ auto-correlation and $r_{\min} = 40 h^{-1}$ Mpc for the $\text{Ly}\alpha$ -QSO cross-correlation. The reason for the more conservative choice in the cross-correlation is due to the fact that we do not have a model to gauge the impact of small-scale non-linearities in the cross-correlation (like we do with the empirical model in [Equation \(9\)](#) for the auto-correlation). Our choice of minimum scale cut for the cross-correlation is the same as the one used for the DESI DR1 analysis in [\[16\]](#), while for the auto-correlation it is slightly more conservative (it was $r_{\min} = 25 h^{-1}$ Mpc in [\[16\]](#)). In terms of the maximum scale cut, we chose $r_{\max} = 200 h^{-1}$ Mpc, which is the largest scale included in our measurement. In principle, one could also include scales larger than that, but the significant increase in the size of the covariance matrix and diminishing returns when it comes to AP constraints precluded us from at-

tempting it.

We also found that our baseline model provides a good fit to the mock correlations (see [Figure 3](#) in [H26](#)). Based on the quality of the fit and the fact that our baseline model produces unbiased AP constraints given our scale cuts, we conclude that our baseline model and analysis configuration have been validated.

Further tests on AP using mocks are presented in [H26](#), including statistical tests of the mock population. [H26](#) shows that our AP uncertainties are consistent with the distribution of results in individual mocks, which validates the statistical uncertainties of our measurements. Finally, Appendix D of [H26](#) shows results from a subset of mocks with a few different fiducial cosmologies, including variations in ΛCDM and $w_0 w_a\text{CDM}$ parameters, and finds no significant impact on Broadband AP constraints from the choice of fiducial cosmology.

To conclude this section, we briefly discuss BAO results from our mocks. In [Figure 3](#), we show BAO results from our full-shape fits in mocks.¹⁶ We find a small but significant bias present in both **CoLoRe 2LPT** and **AbacusSummit** mocks, primarily in the AP direction (ϕ_p). The results from the four types of **AbacusSummit** mocks are not fully consistent with each other in terms of α_p , but they all indicate a fairly consistent bias in ϕ_p of $\sim 0.8\%$. The primary cause of the bias in **AbacusSummit** mocks is likely the BAO shift due to non-linear growth as discussed in [\[73\]](#). The **CoLoRe 2LPT** bias is broadly consistent in direction and magnitude with the biases in **AbacusSummit** mocks. These results indicate that an extra systematic uncertainty is needed to account for this bias in the absence of an analytic model to correct it (for detailed discussions on this, see [\[72, 73\]](#)). Indeed, a systematic uncertainty was added to the DESI DR2 BAO measurement to account for the expected bias arising from this BAO shift [\[14\]](#). We revisit this systematic uncertainty in light of the new results here in [Section V B](#) below.

B. Blinded robustness tests

We conducted numerous tests on blinded data. These include several data splits and a large number of analysis variations. The goal of the data splits is to check that different subsets of the data agree with each other, while the goal of the analysis variations is to test the robustness of our measurements against different choices made in our analysis. The full set of blinded data tests is described in detail in [H26](#). Here, we give an overview of the most important tests.

Two of the data splits are shown in [Figure 4](#). The first is a cut in the signal-to-noise ratio (SNR), defined

¹⁶ BAO results are fairly insensitive to the minimum scale cut used, so we only show results using our baseline model setup here.

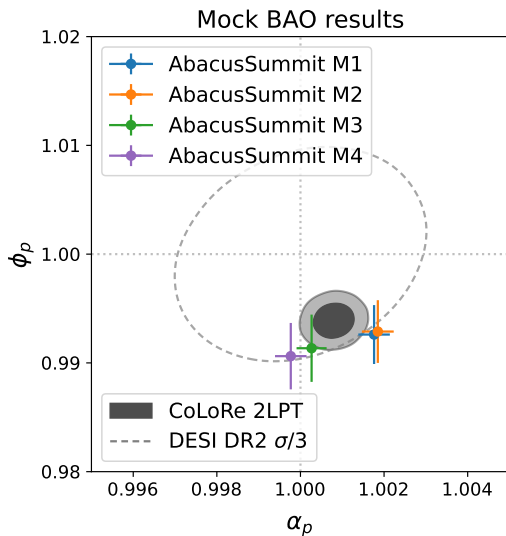


FIG. 3. BAO constraints in mocks. The gray dashed contour indicates the $1/3 \sigma_{\text{DR2}}$ threshold, while the dotted cross indicates the truth in mocks. We show results from fits to the four different versions of **AbacusSummit** mocks as points with error bars, and from the stacked correlations of 400 **CoLoRe 2LPT** mocks as the black contours. These results indicate a small but significant BAO bias that is fairly consistent between the different mock versions. We account for this bias in our systematic error budget, and do not adjust the measurement.

as the mean SNR in the rest-frame wavelength range, $1420 < \lambda_{\text{rest}} < 1480 \text{ \AA}$ [13]. The High SNR sample are quasars with $\text{SNR} > 4.25$, while the Low SNR sample are those with $\text{SNR} < 4.25$. The threshold was chosen such that the two samples have roughly similar uncertainties on their measured correlations. Many effects could be different in high versus low SNR spectra, with perhaps the most important ones being our ability to detect and mask contaminants such as DLAs and BALs [40, 75], as well as the extent to which noise impacts the quasar continuum fitting. Figure 4 shows good agreement between the results from the two SNR splits.

The second data split shown in Figure 4 is between the two $\text{Ly}\alpha$ auto-correlations and the two $\text{Ly}\alpha$ -QSO cross-correlations. This is a very important test to pass because the auto and cross-correlations are affected differently by most of the important contaminants and there are several systematic effects that are only present in one and not the other. These include the sky contamination for the auto-correlation, and the redshift errors and transverse proximity effect for the cross-correlation. Furthermore, the models for HCD and metal contamination, UV background fluctuations, small-scale non-linearities, and the distortion due to continuum fitting are all different to some extent between the auto- and cross-

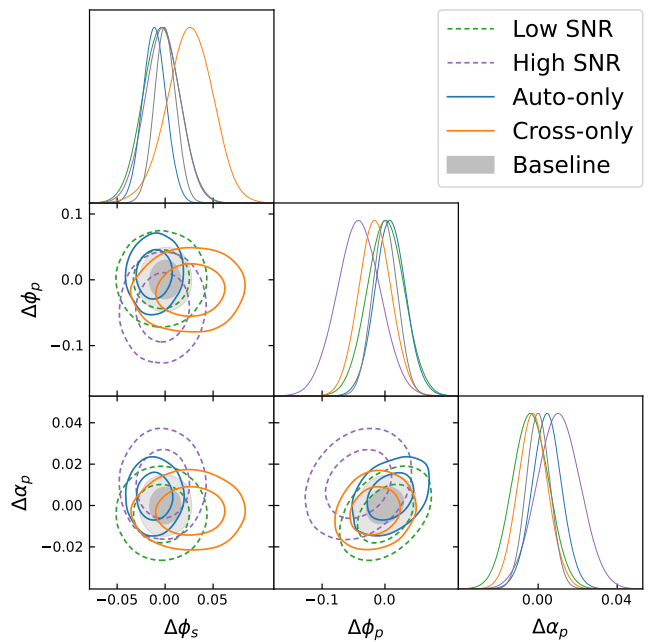


FIG. 4. Comparison of broadband AP and BAO results from data splits. We show results from the low- (high-) SNR split using green (purple) dashed lines. The constraints from the two $\text{Ly}\alpha$ auto-correlations in blue are compared against the constraints from the cross-correlations in orange (orange). Both data splits produce results that are consistent with each other and with the baseline constraints shown in gray.

correlation.¹⁷ As before, Figure 4 shows good agreement between the auto-correlations and the cross-correlations.

We conducted additional data splits based on the A and B regions of the $\text{Ly}\alpha$ forest, into north and south samples based on the imaging surveys used for targeting, and based on the equivalent width of the CIV emission line. These splits are presented in H26 and show good agreement with each other.

We also performed a wide range of analysis variations, testing our choices at each level of the analysis. The first three sets of tests include variations in the data set and catalogs, variations in the continuum fitting process, and variations in the measurement of the correlation functions and covariance matrix. We largely followed previous DESI $\text{Ly}\alpha$ forest analyses in the choices of tests to run in these categories. Notable tests in these categories include changes to the rest-frame wavelength cut, the SNR cut used to mask DLAs, the minimum length of forests, the binning in observed wavelength and in co-moving coordinates, the number and size of HEALPix regions used to estimate the covariance matrix, and a number of variations in the configuration of the distortion matrix computation. All of the tests produce results

¹⁷ For most of these it is because they only affect the $\text{Ly}\alpha$ forest and not quasars.

within our $1/3\sigma$ threshold from the baseline. For the full set of tests and results, see Section VB of H26.

The final set of variations includes those that change parameter priors and the model for the correlation functions. These are fairly distinct from the tests done for BAO analyses because the goal is to test the robustness of the constraint on the broadband AP parameter. First, a number of our nuisance parameters, which are not well constrained by the data, have informative Gaussian priors (see Section B). We tested all of these by either removing the prior or using a different prior to gauge whether the choice of prior impacts our measurements. In all cases, the results are within our threshold. Secondly, we tested freeing a number of parameters that are fixed in our baseline model. In particular, marginalizing over all parameters in the empirical model for small-scale non-linearities in Equation (9) does not significantly impact our measurements and uncertainties. We have also tested removing this correction entirely (i.e., only relying on linear theory) and again found no significant impact on our constraints. This reinforces our conclusion that the analysis configuration primarily targets the larger scale, linear regime.

We tested changing the damping that accounts for redshift errors from a Lorentzian to Gaussian, removing the model for UV background fluctuations, and either fixing the isotropic scale parameter for the smooth component, $\alpha_s = 1$, or allowing two different α_s parameters for the auto- and cross-correlation. In all of these cases, the results are within $1/3\sigma$ of our baseline measurement. Finally, we have tested that masking out the first $r_{\parallel}$ bin (i.e., imposing a cut $r_{\parallel} > 4 h^{-1}$ Mpc) does not have a significant impact. This is particularly relevant for the sky contamination, which is entirely contained within that first $r_{\parallel}$ bin before the distortion due to the continuum fit (Section III B and [60]).

C. Discussion

In this section, we discuss our validation results and provide an overview of aspects of our analysis and mocks that are most relevant for these results. Our goal in this analysis was to target large linear scales in the Ly α forest auto-correlation and its cross-correlation with quasars. For the auto-correlation, we use an empirical model to gauge the impact of small-scale non-linearities (Section III B). However, when using the small-scale marginalization approach introduced in Section III C, together with the scale cuts discussed above, we find that our AP measurement is insensitive to the presence and parameter values of this model (Section IV B). This confirms that we are not sensitive to small-scale non-linearities, at least in the auto-correlation. On the other hand, for the cross-correlation, we use a simple linear theory model. Because of this, we chose more conservative scale cuts relative to the auto-correlation. We have also shown consistent AP results between the auto- and

cross-correlation (Section IV B) and validated our measurements using mocks with realistic quasar clustering (Section IV A) based on the AbacusSummit simulations. This approach works well at the precision of DESI DR2, given the threshold used in our validation. However, its success was not guaranteed, and indeed we expect this approach to encounter a systematic floor as constraints become increasingly precise. Therefore, future analyses using larger data sets, or those attempting to access smaller scales in the Ly α correlations or power spectra [76], are likely to require either an effective field theory approach [77–79] or an emulator approach [e.g. 50].

A major aspect of our validation centers on using DESI DR2 mocks to demonstrate that we can recover the true cosmology with our baseline analysis. Therefore, we also discuss the level of realism in these mocks and whether the approximations used are likely to impact our results. First, there are a number of smaller effects that we know are present in real data but are not present in the mocks. These include CIV metal contamination, the transverse proximity effect, correlated noise due to sky contamination, and UV background fluctuations. We found that ignoring UV background fluctuations and the transverse proximity effect by removing our model for these effects does not significantly affect the AP constraint (see Section VB in H26). On the other hand, ignoring the sky contamination does have a significant impact (see Appendix C3 in H26). As this effect mostly contributes to the $r_{\parallel} < 4 h^{-1}$ Mpc bins in the auto-correlation [60], we performed a test that masks out these bins, thereby removing most of the sky-contamination effect. This did not have a significant impact on our AP constraint (H26), which validates our current model for this effect (Section III B). Finally, for the CIV metal contamination, we currently use an informative prior on the linear bias parameter for CIV absorption. This prior is based on measurements by [60] using DESI CIV forest data and is necessary because Ly α forest correlations cannot constrain this effect. However, the prior we use is already a factor of three wider than the measurement of [60], and the CIV bias parameter does not display a significant correlation with ϕ_s . These effects are discussed in more detail in H26.

Next, we turn our attention to effects which are modeled in mocks, but are not necessarily as realistic as they could be. This includes the most important Ly α forest contaminants: HCDs and metals. HCDs are introduced in mocks in a similar fashion to quasars, by sampling the peaks of the density field [28], but their clustering and especially their column density distribution function are fairly uncertain. However, we find that they have a significant systematic impact on our correlation only at scales smaller than those we fit. When using scales $r > 25 h^{-1}$ Mpc we measure an HCD linear bias consistent with 0 at 68% confidence, whereas when we include $r < 25 h^{-1}$ Mpc we measure a nonzero value at very high significance ($> 5\sigma$). Therefore, we conclude that HCDs are not a significant source of concern given the current precision and scale cuts. In terms of metals, they

are simulated in our mocks by rescaling $\text{Ly}\alpha$ forest flux skewers [28, 80], which means their small-scale clustering is the same as that of the $\text{Ly}\alpha$ forest in mocks, but that is likely not the case in real data [e.g., 81]. To gauge the potential impact of this effect, we allowed greater freedom in the metal- $\text{Ly}\alpha$ cross-correlations¹⁸ model at small scales, using the empirical model in Equation (9) with all parameters free, and found no significant impact on AP. Finally, both our sets of mocks rely on the fluctuating Gunn-Peterson approximation (FGPA, [69]) to compute $\text{Ly}\alpha$ forest fluctuations from the matter density field. The **AbacusSummit** mocks we use have four versions (labeled M1 to M4), corresponding to different parameters and methods for applying the FGPA, and reproduce $\text{Ly}\alpha$ forest 1D and 3D power spectra from hydrodynamical simulations to various degrees of realism [67]. Therefore, an important part of our validation was to show that all four versions of the **AbacusSummit** mocks produce unbiased constraints within our thresholds, which indicates that our results are not very sensitive to how the FGPA is used. These results are shown in Figure 2 and discussed in Section IV A.

Based on the tests presented and discussed above, along with those presented in H26, we concluded that our Alcock–Paczynski measurements are robust and proceeded to unblind our results. Because our validation with mocks produced broadband AP results consistent with the truth within our threshold, combined with the fact that our blinded data tests showed consistent and robust constraints on ϕ_s , we decided not to include an extra systematic uncertainty for this parameter.

V. RESULTS

We next turn our attention to the main results of this work. We begin this section by going through the model fits to our correlation function and present the main parameter constraints in Section V A. After that, we discuss the systematic error budget on the BAO parameters in Section V B. Finally, we present our cosmological distance constraints in Section V C.

A. Alcock–Paczynski measurements

We use the **Vega** package to compute the model and perform the inference. The likelihood is Gaussian, and we sample the full posterior distribution using the Nested Sampler **PolyChord** [82, 83]. To obtain the best fit model we use the **iminuit** minimizer [84, 85]. The effective number of data points is 11388, computed by subtracting the number of analytically marginalized parameters, 402,

from the total number of fitted data points, 11790 (Section II). The measured correlations and best-fit model are shown using the wedge compression in Figure 5 and using the shell compression in Figure 6. The model fit is good, with a $\chi^2/(N_{\text{data}} - N_{\text{param}}) = 11492.0/(11388 - 21) = 1.011$, and PTE = 0.20. Because we effectively implement the scale cuts in the undistorted model via small-scale marginalization and then fit all data points down to zero, the model perfectly fits the data below r_{min} , but those points do not meaningfully contribute to our parameter constraints.

Figure 5 also shows the best-fit model from the DR2 BAO analysis for comparison. While the scale cuts are slightly different ($r_{\text{min}} = 30 h^{-1}$ Mpc for the cross-correlation, versus $40 h^{-1}$ Mpc here), this is a good display of how the changes in the analysis between BAO and full-shape impact the resulting best-fit model. In particular, the model appears to be significantly improved in the transverse wedge ($\mu < 0.5$) and around the BAO peak in the two intermediate wedges.

Figure 6 shows the anisotropy present in our correlations and how well it is fit by the model. This figure also includes the residuals for each shell, which are consistent with our expectations. There are no significant outliers or patterns that could hint at biases in the measured anisotropy. When interpreting the wedge and shell plots, it is important to note that there is significant cross-covariance between neighboring points, which can give rise to apparent features due to noise. Outliers in these measured correlations were studied in section IVA of the DR2 BAO paper [14] using the BAO model. They concluded that there is no evidence of significant disagreement between the data and their best-fit model. This conclusion carries over to our analysis as well, given that our model improves upon theirs.

Our main measurements are summarized in Table II, where we focus on the broadband AP measurement and the BAO constraints. The full set of results, including nuisance parameters, can be found in Section B. The primary new result of this work is the broadband AP constraint, $\phi_s = 1.007 \pm 0.011$, obtained from the smooth component of the $\text{Ly}\alpha$ forest correlations. This represents a 1.1% constraint on the distance ratio D_M/D_H at $z_{\text{eff}} = 2.33$, which corresponds to a factor of ~ 1.8 improvement with respect to the constraint from the DR2 BAO analysis (2% constraint on AP; [14]) and a $\sim 45\%$ improvement with respect to the DR1 full-shape analysis (1.6% constraint on AP; [16]).

In Table II we also report the BAO measurements obtained from our full-shape fit, which include the AP parameter measured from the BAO feature, $\phi_p = 1.001 \pm 0.020$, and the isotropic BAO scale, $\alpha_p = 1.001 \pm 0.006$. The AP measurement from BAO is consistent with our new broadband AP result, which represents another useful consistency check. This is because the two are constrained by different components of the correlation function and are fairly independent, with a correlation coefficient $\rho < 1\%$. Our final results will represent the com-

¹⁸ Which are the primary source of this contamination.

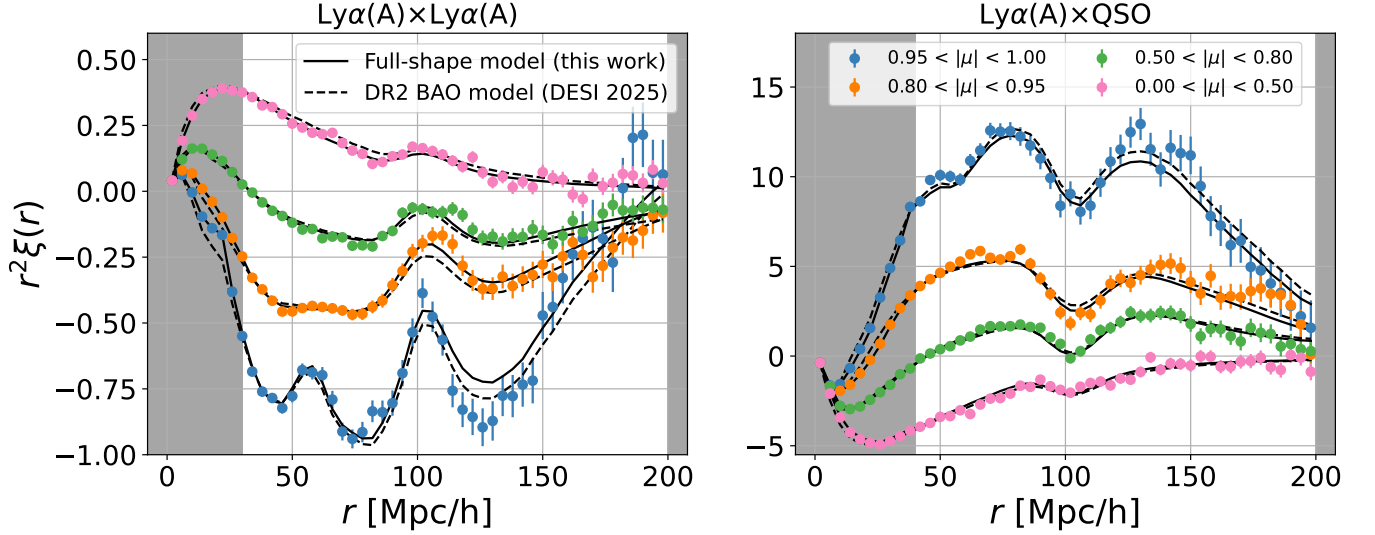


FIG. 5. Wedge compression of the $\text{Ly}\alpha$ auto-correlation (left) and its cross-correlation with quasars (right). The black lines show the best-fit models, with the continuous line indicating the full-shape model described here and the dashed line showing the best-fit model from the DR2 BAO analysis. Gray bands indicate the scale cuts for the full-shape analysis. While the scale cuts differ slightly between the two fits, this provides a good representation of the evolution of the model.

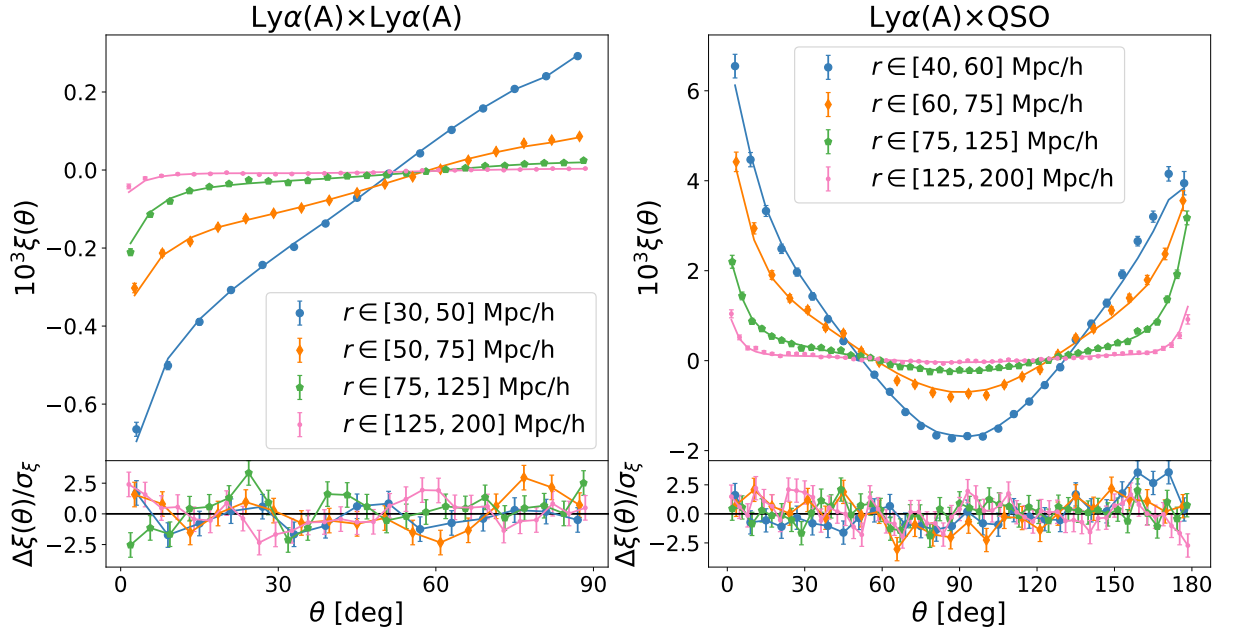


FIG. 6. Shell compression of the $\text{Ly}\alpha$ auto-correlation (left) and its cross-correlation with quasars (right), along with the best fit model (top) and residuals (bottom). These figures display the anisotropy across four radial shells, with the most relevant ones for our analysis being the smaller-separation shells in blue and orange.

bination of the broadband AP and BAO measurements, and are described in [Section V C](#) below. As discussed in [Section IV C](#), we did not include an extra systematic uncertainty on ϕ_s . However, following DESI DR2 BAO, we do include a systematic uncertainty in the BAO measurements, which are described in the next section.

B. BAO systematic uncertainties

At the statistical precision of DESI DR2, BAO constraints from the $\text{Ly}\alpha$ forest require an assessment of possible shifts in the BAO peak position caused by non-linear evolution. Such shifts can arise when the fitting model does not fully capture the relevant non-linear ef-

Description	Distance ratio	Measurement
Broadband AP	D_M/D_H	$\phi_s = 1.007 \pm 0.011$
AP from BAO	D_M/D_H	$\phi_p = 1.001 \pm 0.020$
Isotropic BAO	$(D_M D_H)^{1/2}/r_d$	$\alpha_p = 1.001 \pm 0.006$

TABLE II. Main DR2 Ly α forest full-shape results. We report the broadband AP constraint (the primary new result of this work), together with the AP and isotropic BAO constraints from the same dataset. For each parameter, the middle column indicates the corresponding distance ratio that is constrained by that parameter. Parameter values in the final column correspond to the ratio of these distances with respect to their fiducial values.

fects [72, 73, 86]. In the DR2 BAO analysis, this contribution was treated as a theoretical systematic uncertainty on the BAO distance parameters, rather than as a direct correction to the measured values. This choice is motivated by the fact that no reconstruction is applied to the Ly α forest field, so residual BAO shifts may remain non-negligible at sub-percent precision [14].

Section VIA of the DR2 BAO paper reviews recent Ly α BAO-shift studies based on the **AbacusSummit** mocks and on effective field theory (EFT) modeling calibrated to hydrodynamical simulations. These studies broadly indicate shifts of order a few $\times 10^{-3}$, but they differ in both magnitude and sign, and each currently has only moderate statistical significance. Given this lack of convergence, the DR2 BAO analysis adopted a conservative covariance-level systematic term, with $\Delta\alpha_{\parallel} = 0.3\%$ and $\Delta\alpha_{\perp} = 0.3\%$, added in quadrature to the statistical covariance, and reported both statistical-only and statistical+systematic constraints separately.

Given the results we presented in Section IV A, we decided to take a slightly different approach for our BAO systematic error budget. As we saw, both **AbacusSummit** and **CoLoRe 2LPT** mocks indicate a systematic uncertainty that is significantly larger in the AP direction (ϕ_p) than that used in the DR2 BAO paper. Therefore, we decided to update the BAO systematic error budget here to account for this fact.

We follow [14] and add the systematic uncertainties to the 2×2 covariance matrix of the BAO parameters, but use the α_p and ϕ_p directions (as opposed to $\alpha_{\parallel}$ and $\alpha_{\perp}$):

$$C(\alpha_p, \phi_p) = C(\alpha_p, \phi_p)_{\text{stat}} + C(\alpha_p, \phi_p)_{\text{sys}}, \quad (20)$$

with

$$C(\alpha_p, \phi_p)_{\text{sys}} = \begin{bmatrix} \Delta\alpha_p^2 & 0 \\ 0 & \Delta\phi_p^2 \end{bmatrix}, \quad (21)$$

where we choose the systematic uncertainties in the isotropic BAO and AP directions to be:

$$\Delta\alpha_p = 0.15\%, \quad (22)$$

$$\Delta\phi_p = 0.8\%. \quad (23)$$

This corresponds to greater uncertainty in the AP direction compared to the DR2 BAO analysis, and smaller uncertainty in the isotropic BAO direction.

We note that the exact magnitude and direction of the BAO shift are still fairly uncertain, as we do not know how close to reality the shift observed in our two sets of mocks is. The fact that **CoLoRe 2LPT** and **AbacusSummit** mocks give consistent results represents an important step forward in our understanding of this effect, but we note that predictions based on EFT fits in **AbacusSummit** mocks and hydrodynamical simulations still disagree on the exact direction of the shift [72, 73]. Due to this disagreement, we chose not to adjust our results by the shift measured in Section IV A. However, both our results and those of [72, 73] indicate that the shift is at most at the level of about 1/3 of our current statistical uncertainty in BAO. This means that the systematic error and the corresponding systematic uncertainty used here have a small impact on the resulting BAO constraints. Finally, the broadband AP constraint presented here is significantly more precise than the BAO measurement, further diluting the importance of this choice.

C. Cosmological distances

The primary result presented here is based on the broadband AP signal (ϕ_s) measured using the smooth component of the Ly α forest correlation functions. This provides a tight constraint on the distance ratio D_M/D_H at $z_{\text{eff}} = 2.33$, given by:

$$\text{broadband AP : } D_M/D_H(z_{\text{eff}}) = 4.578 \pm 0.052. \quad (24)$$

When combining this with the BAO AP constraint (ϕ_p), including the systematic uncertainty discussed above, we measure the AP signal from the full shape of Ly α correlations:

$$\text{full shape AP : } D_M/D_H(z_{\text{eff}}) = 4.572 \pm 0.046, \quad (25)$$

which corresponds to a 1.0% constraint of the Alcock–Paczynski effect.

Our final results are given by the combination of the full-shape AP and isotropic BAO constraints (including the systematic uncertainty). This means we are using all the information from our three main parameters: ϕ_s , ϕ_p , α_p . The corresponding distance constraints are:

$$\begin{array}{l} \text{DESI Ly}\alpha \\ \text{full-shape} \end{array} \left\{ \begin{array}{ll} D_M(z_{\text{eff}})/r_d & = 39.32 \pm 0.33, \\ D_H(z_{\text{eff}})/r_d & = 8.600 \pm 0.066, \\ \rho(D_M/r_d, D_H/r_d) & = 0.225, \end{array} \right. \quad (26)$$

where ρ is the correlation coefficient between the two parameters. These results correspond to a 0.84% constraint on D_M/r_d , and a 0.77% constraint on D_H/r_d .

The DESI instrument was designed to achieve combined precisions of about 0.9% in both D_M/r_d and

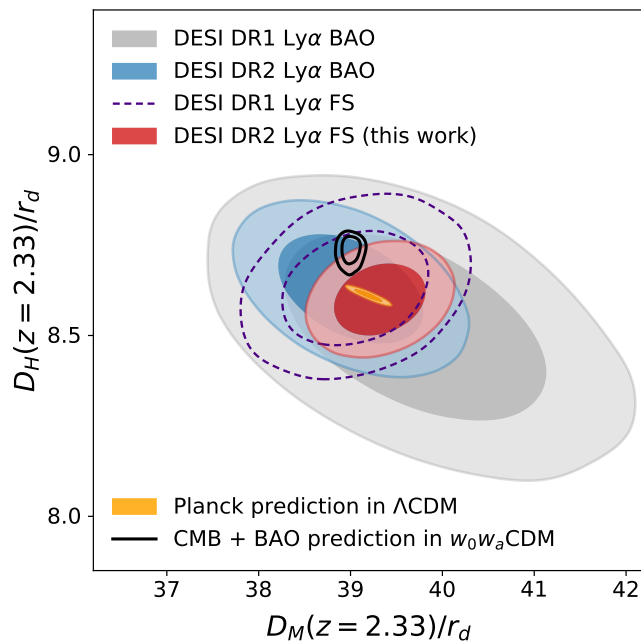


FIG. 7. Constraints on the distance pair $(D_M/r_d, D_H/r_d)$ at $z_{\text{eff}} = 2.33$ from Ly α forest measurements, shown as 68% and 95% confidence contours. The red filled contours give the DESI DR2 full-shape result from this work, compared with DESI DR2 BAO (blue), DESI DR1 full-shape (purple dashed), and DESI DR1 BAO (gray). The orange contour shows the Planck CMB prediction in Λ CDM, while the black contour shows the CMB+BAO prediction in w_0w_a CDM. The Ly α full-shape results provide the tightest distance constraints from large-scale structure at $z > 1$, and the uncertainty is now small enough to begin to separate between Λ CDM and w_0w_a CDM predictions from other data sets in this space.

D_H/r_d in the redshift range of the Ly α forest ($1.9 < z < 4.0$) from the original 5-year survey [2, 19]. Those forecasts were based on BAO-only, as no Ly α forest 3D full-shape analyses had been performed at the time. Therefore, with the new constraints presented here, we have already exceeded the original DESI requirements using only the first 3 years of DESI data.

We compare our results with previous DESI Ly α forest measurements in Figure 7. The first DESI results from DR1 BAO are shown in gray. When they were released two years ago, they were the state-of-the-art distance constraints in the redshift interval $1 < z < 4$, narrowly improving over the previous best constraints from SDSS [12]. Since then, DESI has also released DR1 Ly α forest full-shape measurements (dashed purple) and DR2 BAO measurements (blue), both of which significantly improved these distance constraints and were in excellent agreement with each other. The measurement we have presented here (red) provides yet another significant improvement and corresponds to a 6.7-fold reduction in the area of the contours compared with the original DESI DR1 BAO constraint.

The precision gain from the new measurement allows us to begin to separate between the Λ CDM and w_0w_a CDM predictions of $D_M(z_{\text{eff}})/r_d$ and $D_H(z_{\text{eff}})/r_d$ from other cosmological tracers at more than 1σ . This is illustrated in Figure 7, where we also show the Λ CDM prediction from Planck (orange) and the w_0w_a CDM prediction from the joint CMB¹⁹ and DESI BAO analysis (black).²⁰ The DR2 Ly α forest constraints are in excellent agreement with the Λ CDM prediction, but show a roughly 2σ tension with the w_0w_a CDM prediction. We will investigate this tension and its implications in the following Section.

To summarize, we have provided our final measurements in Equation (26), which represent significant improvements in the precision of cosmological distances at $z = 2.33$. These measurements allow us for the first time to meaningfully distinguish among different state-of-the-art dark energy constraints with the Ly α forest alone.

VI. COSMOLOGICAL CONSTRAINTS

In this section, we discuss the cosmological implications from the Ly α forest full-shape measurements and define the datasets that are used to perform joint constraints on cosmological parameters.

A. Other data sets

We complement the high-redshift information accessible to the Ly α forest by including BAO measurements from galaxy and quasar clustering. We use the DESI DR2 BAO data presented in [5]. This dataset includes measurements of D_V/r_d at $0.1 < z < 0.4$ for the Bright Galaxy Sample (BGS), joint measurements of D_M/r_d and D_H/r_d for the Luminous Red Galaxies (LRG) at $0.4 < z < 0.6$ and $0.6 < z < 0.8$, the combined LRG and Emission Line Galaxies (ELG) tracer at $0.8 < z < 1.1$, the ELG at $1.1 < z < 1.6$, and the Quasar (QSO) clustering at $0.8 < z < 2.1$. For comparison, we also include the Ly α BAO measurements from [14], when necessary. Throughout, we refer to this combination as DESI BAO, and to the combination with the new AP Ly α measurement as either DESI BAO+Ly α FS or simply as DESI.

We also include external data to DESI to constrain parameters beyond the late-time expansion history. In particular, anisotropies in the CMB provide a wealth of information that can help constrain the complete set of base cosmological parameters. Following the previous main DESI analyses, we use data from the *Planck* satellite [87, 88] based on the PR4 release that uses the

¹⁹ See Section VIA for the combination of CMB results used here.

²⁰ The prediction from the joint w_0w_a CDM analysis that also includes Supernovae is not meaningfully different, as SNe largely contribute to improvements at lower redshifts ($z < 1.5$).

NPIPE maps [89]. Additionally, we now include new data from ground-based CMB experiments. In particular, we include the Atacama Cosmology Telescope (ACT) data release 6 (DR6) [90–92] and the South Pole Telescope (SPT)-3G D1 [93]. We combine the three CMB surveys to measure temperature (TT), polarization (EE), and temperature-polarization cross (TE) spectra, as follows. For the low- ℓ range ($\ell < 30$) we use the **Commander** likelihood for TT [94], and the **Srol12** likelihood for EE [95], both based on *Planck*. For the high- ℓ range ($\ell \geq 30$), the CMB surveys have overlapping measurements and we choose to truncate the high- ℓ CMB likelihoods before combining them to avoid the need to model cross-correlations between CMB surveys. We also use foreground-marginalized versions of the CMB likelihoods to avoid leaving foreground nuisance parameters unconstrained by this truncation. We use the lite-version of the **CamSpec** TTTEEE likelihood [96] as presented in [97] over the range $[(\ell < 1500, TT), (\ell < 1000, TE), (\ell < 600, EE)]$ and complement it with the ACT-lite likelihood [90] evaluated from that multipole range up to $\ell < 6500$ (see also [98] for an alternative truncation scheme). Given the small footprint of SPT-3G D1, cross-correlations with the other CMB surveys are less relevant, such that the data can be combined without truncation. We use the SPT-lite likelihood as presented in [93]. Additionally, we include information from the 4-point lensing reconstruction and use the unified CMB lensing likelihood based on *Planck*, ACT, and SPT presented in [99].

Finally, we include Type Ia supernovae data from the recently reanalyzed DES-Dovekie sample [100], which now comprises a sample of 1820 SNe (197 SNe at $z < 0.1$ and 1623 SNe at $0.1 < z < 1.13$). This sample has 1718 SNe in common with the previous DES sample originally presented in [101] and around 100 SNe that were not previously included. The differences between the two samples are a result of new light-curve fits after updates to the calibration, SALT model, re-run simulations, and other changes as described in [100, 102]. Overall, this new SNe sample led to a $\sim 1\sigma$ shift towards a smaller Ω_m value in Λ CDM compared to the original DESY5-SNe sample. The level of agreement with other SNe datasets, such as Pantheon+ [103] and Union3.1 (originally Union3 [104]), has improved significantly in the recent updated analysis [105]. Given this agreement, we only present results for the DES-Dovekie sample, as it provides the tightest constraints, and defer combinations with other SNe samples to future work.

Finally, to calibrate the sound horizon, r_d , we also use a measurement of the baryon density, $\Omega_b h^2 = 0.02218 \pm 0.00055$, from [106]. This is based on measurements of the primordial deuterium abundance [107] and Big Bang Nucleosynthesis (BBN). When combining with this measurement, we will refer to it as BBN.

We perform cosmological likelihood calculations using the **cobaya** package [108] for Markov Chain Monte Carlo parameter sampling. The priors on the cosmological parameters are identical to the ones used in [5]. We use

a Gelman-Rubin convergence criterion of $R - 1 < 0.01$ [109]. Distances and CMB spectra are computed using the Einstein-Boltzmann solver **camb** [110]. We evaluate the quality of the fits in terms of the χ^2_{MAP} at the maximum a posteriori (MAP) points of each chain, which are found using the **bobyqa** algorithm [111–113].

B. Matter era distance

The Ly α forest enables accurate distance measurements well into the matter-dominated era. This provides an excellent opportunity to test Λ CDM at an epoch, $z_{\text{eff}} = 2.33$, when the contribution of dark energy to the cosmic energy density is greatly suppressed. The distances of Equation (26) can be directly compared with the Λ CDM predictions derived from DESI BAO measurements of galaxy and quasar clustering at lower redshifts. The Ly α measurements are consistent with these predictions, though with a 1.4σ offset from the prediction for the transverse comoving distance of $D_M(z_{\text{eff}})/r_d = 38.82 \pm 0.16$. Combining DESI BAO with Ly α FS improves the Λ CDM constraint by 30% to $D_M(z_{\text{eff}})/r_d = 38.85 \pm 0.11$. However, unlike Equation (26), the inferred distance is not independent of assumptions about dark energy at lower redshifts. Allowing for dynamical dark energy and adopting the CPL parameterization for an evolving dark energy equation of state [114, 115], $w(a) = w_0 + w_a(1 - a)$, we find a constraint of $D_M(z_{\text{eff}})/r_d = 38.91 \pm 0.12$, which corresponds to a 0.5σ increase compared to Λ CDM.

Once combined with acoustic scale information from the CMB, these constraints on $D_M(z_{\text{eff}})/r_d$ provide a useful diagnostic for physics at higher redshifts [116, 117]. Similar to the BAO feature in the Ly α forest, the acoustic peaks in the power spectra of CMB temperature and polarization anisotropies yield a precise determination of the transverse comoving distance $D_M(z_d)/r_d \equiv 1/\theta_d^{\text{CMB}}$, where $z_d \approx 1060$ is the redshift at the drag epoch. Taking the difference with the constraint on $D_M(z_{\text{eff}})/r_d$ from DESI defines a long baseline during the matter era, $D_M(z_d, z_{\text{eff}})/r_d \equiv D_M(z_d)/r_d - D_M(z_{\text{eff}})/r_d$, known as the matter era distance interval [117].

Figure 8 shows constraints on $D_M(z_{\text{eff}})/r_d$ and $D_M(z_d, z_{\text{eff}})/r_d$ for various data combinations and models. The top row shows constraints that are directly derived from Equation (26) and a broad Gaussian prior on θ_d^{CMB} .²¹ This approach is insensitive to assumptions

²¹ Constraints on θ_d^{CMB} are model dependent. For instance, uncertainties increase significantly when varying the sum of neutrino masses. In order to account for this model dependence, we adopt a prior $\mathcal{N}(1.06042, 0.0028^2)$, with a width 10 times greater than the uncertainty of the marginalized constraint from the baseline CMB likelihood, $\theta_d^{\text{CMB}} = 1.06042 \pm 0.00028$, assuming Λ CDM. We refer to this prior as $\theta_{d,10}^{\text{CMB}}$.

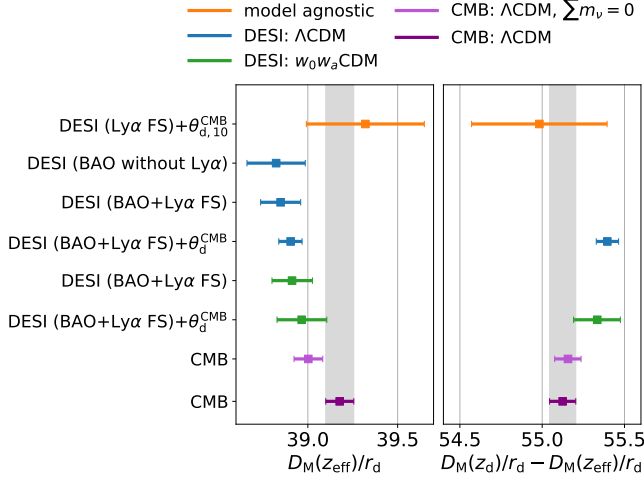


FIG. 8. Constraints on the transverse comoving distance, $D_M(z_{\text{eff}})/r_d$, and the matter era distance interval, $D_M(z_d, z_{\text{eff}})/r_d \equiv D_M(z_d)/r_d - D_M(z_{\text{eff}})/r_d$, for various data combinations and models, showing the posterior means and 68% credible intervals. The top row shows model-agnostic constraints from Ly α FS combined with a broad Gaussian prior on θ_d^{CMB} , with 10 times the uncertainty of that obtained assuming Λ CDM (orange). The next three rows show the geometric constraints for Λ CDM from DESI BAO without Ly α , combined with Ly α FS, and combined with an informative prior on θ_d^{CMB} (blue). This last constraint is in 2.6σ tension with the Λ CDM constraint derived from the full CMB likelihood (purple). Adopting a w_0w_a CDM background reduces the tension (green), as does reducing the sum of neutrino masses (light purple).

about dark energy at lower redshifts, but the uncertainty is currently too large to place meaningful constraints on extensions such as massive neutrinos. The constraints can be tightened considerably by adding information from DESI BAO at lower redshifts, but at the expense of introducing greater model dependence. The fourth row in Figure 8 shows the constraints obtained assuming Λ CDM from Ly α FS combined with DESI BAO at lower redshifts and an informative prior on θ_d^{CMB} . This constraint relies entirely on geometric information and should be confronted with the Λ CDM prediction from the full CMB likelihood, including lensing reconstruction, shown at the bottom of Figure 8. There is a 2.6σ tension between the two constraints, which can be expressed either in $D_M(z_d, z_{\text{eff}})/r_d$ or in $D_M(z_{\text{eff}})/r_d$, since the latter dominates the uncertainty.

These distance measurements offer an alternative lens to analyze joint constraints from DESI and CMB on extended cosmological models. For masses allowed by current data, neutrinos become non-relativistic after recombination, increasing the matter density after some redshift $z_{\text{nr}} < z_d$. This increases the CMB prediction for $D_M(z_{\text{eff}})/r_d$ more than for $D_M(z_d)/r_d$, thus reducing the interval $D_M(z_d, z_{\text{eff}})/r_d$. The late-time geometric constraints from DESI are entirely unaffected, but

once combined with θ_d^{CMB} , the prediction for the interval $D_M(z_d, z_{\text{eff}})$ increases for larger neutrino masses. Larger neutrino masses thus increase the tension in both directions. As a result, combinations of DESI and CMB data yield strong upper limits on the neutrino mass, even pushing towards negative effective neutrino masses if allowed [118–121], in order to compensate for the difference in the matter density [117, 122].

While physical neutrino masses increase the tension with the CMB, dynamical dark energy can partially resolve it. This is best understood in terms of the low-redshift distance ratio $D_M(z_{\text{eff}})/r_d$, since $D_M(z_d)/r_d$ is much less sensitive to low-redshift assumptions about dark energy. The geometric constraints from DESI, combined with θ_d^{CMB} , define a sharp degeneracy direction in the w_0 - w_a plane (see Section VID). Moving away from Λ CDM along this degeneracy increases the DESI constraint on $D_M(z_{\text{eff}})/r_d$ by 0.5σ . At the same time, moving in that direction reduces the CMB prediction for $D_M(z_{\text{eff}})/r_d$, alleviating the tension and the need for smaller neutrino masses.

C. Λ CDM results

We next turn our attention to results in the flat Λ CDM model. When using only DESI data, the parameters we can constrain are the matter fraction, Ω_m , and the product of the Hubble constant and the size of the acoustic scale, $H_0 r_d$ [5]. When inferring these quantities from the Ly α forest results alone, we find:

$$\left. \begin{aligned} \Omega_m &= 0.325 \pm 0.018, \\ hr_d &= (97.9 \pm 2.1) \text{ Mpc}, \end{aligned} \right\} \text{ Ly}\alpha \text{ FS} \quad (27)$$

with a correlation coefficient of $\rho = -0.96$ and $h = H_0/(100 \text{ km s}^{-1} \text{ Mpc}^{-1})$. These results are consistent with the DESI BAO measurements at lower redshifts, as illustrated in the top panel of Figure 9. When compared to the combined DESI BAO constraints excluding Ly α , we find them statistically consistent within 1σ (orange versus green in Figure 9). Also, the new Ly α FS constraints represent about a factor of 2 precision improvement in Ω_m with respect to its BAO counterpart. This means that the new Ly α forest result can be safely combined with DESI BAO measurements at $z < 2$ within Λ CDM. The joint DESI results are:

$$\left. \begin{aligned} \Omega_m &= 0.3012 \pm 0.0079, \\ hr_d &= (101.19 \pm 0.68) \text{ Mpc}, \end{aligned} \right\} \begin{array}{l} \text{DESI} \\ \text{BAO+Ly}\alpha \text{ FS} \end{array} \quad (28)$$

with a correlation coefficient of $\rho = -0.91$. These constraints correspond to a 9% improvement in Ω_m and 7% in hr_d when compared to DESI BAO results alone.

Figure 9 shows the updated DESI results (blue), which consist of the combination of Ly α forest full-shape constraints and BAO from galaxy clustering. In the lower

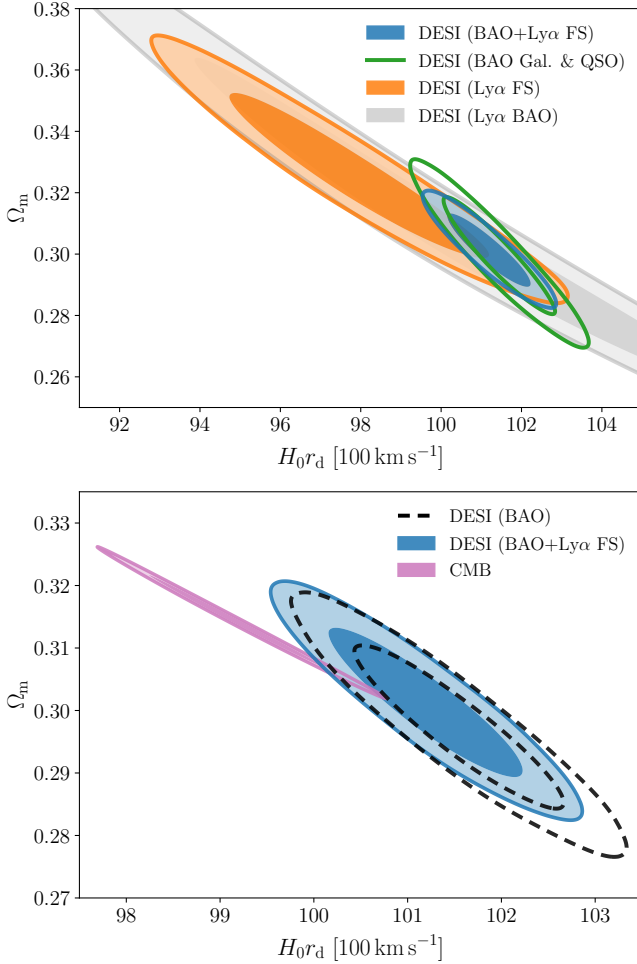


FIG. 9. **Top:** Marginalized 68% and 95% confidence regions representing the posterior constraints on Ω_m and $H_0 r_d$. The light gray and green contours represent the individual constraints from Ly α BAO and BAO from galaxy and quasar clustering, respectively. The orange contour shows the updated constraint from the Ly α FS analysis and the blue contour represents the updated DESI BAO+Ly α FS results. **Bottom:** Constraints comparing the updated DESI results with respect to the CMB prediction in the Λ CDM model. The DESI BAO constraints (same as the DR2 BAO results presented in [5]) are shown as the black dashed contour, while the CMB constraints are shown using the pink contour. Adding the new Ly α forest measurement to DESI BAO improves the combined DESI result by 8% (blue) and leads to a slight shift towards the CMB, reducing the tension between the two.

panel, we compare the new results with the DESI BAO-only constraints (dashed black) and with the CMB constraints (pink). Adding the new Ly α forest measurement leads to slightly tighter constraints as described above, and also to a small shift that brings the DESI result closer to the CMB one. The discrepancy between the

CMB combination used here and DESI BAO is 2.4σ ²² (black versus pink), and this is reduced to 2.2σ if DESI BAO+Ly α FS (blue versus pink) is instead considered. The combination DESI BAO+Ly α FS and CMB leads to $\Omega_m = 0.3042 \pm 0.0033$. In addition, the corresponding spectral index constraint is $n_s = 0.9739 \pm 0.0029$, which is roughly a 0.4σ shift lower compared to the value found by [123] in the context of implications for inflation, using the DESI BAO+ACT combination. Moving forward, we also refer to the combination of DESI Ly α full-shape and galaxy BAO constraints simply as DESI, unless explicitly comparing to DESI BAO.

The discrepancy between CMB and DESI BAO results leads to models with a more flexible background expansion history, such as $w_0 w_a$ CDM, being preferred over Λ CDM [5, 124]. Therefore, at first glance, we expect a reduction in the tension between the CMB and DESI to directly lead to a reduction in the preference for these models compared to Λ CDM. That is indeed what happens when comparing $w_0 w_a$ CDM and Λ CDM, as we will see in the next section. However, the situation is more nuanced, because the new Ly α result is at high redshift ($z = 2.33$), whereas the DESI BAO measurements driving the tension are at low redshift ($z < 1$). This means that Λ CDM might still not be the model that best fits the joint DESI and CMB data. We explore the implications of this in Section VID below.

We next turn to comparisons with results from SNe analyses. The DES Dovekie SNe sample constrains $\Omega_m = 0.330 \pm 0.015$ [102], while the latest constraints using the Union sample are: $\Omega_m = 0.334^{+0.025}_{-0.024}$ [105, 125]. Both of these are in good agreement with the Ly α forest measurement in Equation (27). Notably, the DESI Ly α result alone is 30% tighter than the Union one, and only 20% weaker than the tightest SNe constraint from the DES Dovekie sample. Adding the new Ly α forest constraint to the rest of the DESI BAO measurements results in slightly larger values of Ω_m , reducing the tension with respect to DES Dovekie from 1.9σ ²³ to 1.7σ , and with respect to Union from 1.4σ to 1.3σ .

To measure the Hubble constant, we need to calibrate the BAO ruler by constraining r_d . Assuming standard pre-recombination physics, and a fixed neutrino sector with $\sum m_\nu = 0.06$ eV, the only missing quantity is a constraint on the baryon density $\Omega_b h^2$. Using the $\Omega_b h^2$ prior from BBN [106], we can obtain a constraint on the Hubble constant that is independent of CMB anisotropies. For the Ly α forest full-shape measurement, this corre-

²² Computed using the Gaussian approximation for the two sets of 2-dimensional results. See Eq. 18 of [5].

²³ The tension was originally 2.9σ in [5], when comparing results from DESI BAO and the original DES Y5 SNe sample, and was reduced to 1.9σ with the DES Dovekie sample [126].

Model/Dataset	Ω_m	H_0 [km s ⁻¹ Mpc ⁻¹]	$10^3\Omega_K$	$\sum m_\nu$ [eV]	w or w_0	w_a
ΛCDM						
DESI (Ly α FS)	0.325 ± 0.018	—	—	—	—	—
DESI	0.3012 ± 0.0079	—	—	—	—	—
DESI (Ly α FS)+BBN	0.326 ± 0.018	66.5 ± 1.3	—	—	—	—
DESI+BBN	0.3020 ± 0.0075	68.47 ± 0.57	—	—	—	—
DESI+CMB	0.3042 ± 0.0033	68.11 ± 0.24	—	—	—	—
$w_0 w_a$CDM						
DESI	$0.363^{+0.033}_{-0.017}$	—	—	—	$-0.43^{+0.31}_{-0.14}$	< -1.68
DESI+CMB	$0.343^{+0.019}_{-0.022}$	64.5 ± 1.9	—	—	$-0.54^{+0.19}_{-0.21}$	$-1.39^{+0.61}_{-0.50}$
DESI+CMB+DES-Dovekie	0.3134 ± 0.0052	67.37 ± 0.54	—	—	-0.821 ± 0.054	-0.65 ± 0.20
ΛCDM+Ω_K						
DESI	0.289 ± 0.011	—	51.00 ± 32.0	—	—	—
DESI+CMB	0.3038 ± 0.0034	68.51 ± 0.33	2.10 ± 1.1	—	—	—
DESI+CMB+DES-Dovekie	0.3050 ± 0.0032	68.43 ± 0.30	2.20 ± 1.0	—	—	—
ΛCDM+$\sum m_\nu$						
DESI+CMB	0.3017 ± 0.0033	68.36 ± 0.26	—	< 0.0592	—	—
DESI+CMB+DES-Dovekie	0.3030 ± 0.0033	68.26 ± 0.25	—	< 0.0654	—	—
$w_0 w_a$CDM+$\sum m_\nu$						
DESI+CMB	$0.343^{+0.021}_{-0.024}$	64.5 ± 2.0	—	< 0.166	$-0.53^{+0.21}_{-0.25}$	$-1.42^{+0.78}_{-0.58}$
DESI+CMB+DES-Dovekie	0.3129 ± 0.0054	67.40 ± 0.55	—	< 0.128	-0.827 ± 0.056	$-0.62^{+0.25}_{-0.20}$

TABLE III. Summary of cosmological parameter constraints from DESI Ly α FS and DESI BAO + Ly α FS (labeled as ‘DESI’ here) alone and in combination with external datasets. We quote the marginalized posterior means and the 68% credible intervals for all parameters, except for the sum of neutrino masses, $\sum m_\nu$, for which we quote the 95% upper bound. The unreported $\sum m_\nu$ values here corresponds to a fixed 0.06 eV value.

sponds to:

$$H_0 = (66.5 \pm 1.3) \text{ km s}^{-1} \text{ Mpc}^{-1} \quad (\text{Ly}\alpha \text{ FS+BBN}). \quad (29)$$

This represents a $\sim 2\%$ constraint on the Hubble constant, derived entirely from high-redshift data²⁴ and independent of CMB anisotropies. The measurement is consistent with both CMB and DESI BAO + BBN results, and is 1.7σ away from the tightest cosmic distance ladder result from the Chicago-Carnegie Hubble Program [128], and in 4.3σ tension with the latest results from SH0ES [129]. This tension rises to 4.6σ with respect to the baseline result from the H_0 Distance Network Collaboration [130].

Our full set of parameter constraints is presented in **Table III**, and also includes the combination of DESI+BBN and DESI+CMB results. Both of these are consistent

with the results in [5], but with slightly smaller uncertainties.

D. Extended models

In the following, we perform analyses of extended models beyond Λ CDM, covering evolving dark energy, spatial curvature, and a free neutrino mass.

1. Dark energy

The unknown nature of dark energy, as well as the current trends in cosmology, make the search for constraining the equation of state of dark energy and its redshift dependence of particular interest. We model the equation of state relating the pressure and the energy density of dark energy, $P = w\rho c^2$, with the Chevallier-Polarski-Linder (CPL) parameterization, which uses $w(a) = w_0 + (1 - a)w_a$ where $a = 1/(1 + z)$. It is worth noting that [124] showed that an alternative redshift-dependence of the dark energy equation of state or a non-parametric approach like binning or Gaussian processes converges to

²⁴ The BBN constraints are based on primordial deuterium abundance measurements from DLAs in spectra of high-redshift quasars [e.g., 107, 127].

the same conclusions as the CPL parameterization. We refer to this model in the following as w_0w_a CDM. In particular, a sequence of DESI papers [4, 5, 47] have reported a preference for evolving dark energy from joint analyses with external data. Currently and as shown in Section C, the DESI BAO results show a $\sim 3\sigma$ preference in favor of evolving dark energy once combined with CMB data. On the other hand, further reanalyses on the side of the SNe datasets [100, 105] mitigated discrepancies between different SNe samples, leaving results essentially unchanged regardless of which SNe dataset is chosen, with a preference for evolving dark energy ranging from 3σ to 3.5σ for the DESI+CMB+SNe combination.

For this analysis, the high-redshift measurements of cosmological distances accessible to the Ly α forest allow us to set constraints on the expansion history deep into the matter-domination era. In particular, constraining the AP parameter with sufficient precision at high-redshift can potentially help distinguish between a Λ CDM and a w_0w_a CDM model (illustrated in Figure 13 below). Using the new Ly α measurements, the DESI constraints on w_0 and w_a are now

$$\left. \begin{aligned} w_0 &= -0.43^{+0.31}_{-0.14}, \\ w_a &< -1.68. \end{aligned} \right\} \text{DESI BAO+Ly}\alpha \text{ FS} \quad (30)$$

This result is illustrated in Figure 10 (blue contours).

To compare the fit with respect to Λ CDM, we estimate the $\Delta\chi^2_{\text{MAP}} \equiv -2\Delta\ln\mathcal{L}$ evaluated by maximizing the posteriors of each model and comparing the fit for the w_0w_a CDM model to a w_0w_a CDM model with $w_0 = -1$ and $w_a = 0$. We find that $\Delta\chi^2_{\text{MAP}} = -5.6$ for DESI (BAO+Ly α FS), corresponding to a 1.9σ preference for w_0 and w_a distinct from a cosmological constant. This means that the inclusion of the Ly α FS data has slightly increased the $\Delta\chi^2_{\text{MAP}}$ difference between the two models with respect to DESI BAO constraints. On the other hand, the individual χ^2 values have actually increased, indicating that the fit to the data has deteriorated mildly compared to the fit using DESI BAO alone.

The flexibility of the w_0w_a CDM model allows it to provide a good fit to both DESI measurements at late-times and the new Ly α forest measurements at $z > 2$. However, the situation changes when also considering CMB measurements of the acoustic scale at recombination. The acoustic scale is defined as $\theta_* = r_*/D_M$ and its measurement sets an anchor at high redshift, roughly at $z_* \simeq 1090$. This high-redshift anchor removes some of the freedom in the fit for the dark energy parameters, forcing the fit to not deviate too far from the *Planck* value at recombination (although we notice that the acoustic scale inferred by DESI is consistent with the CMB measurement). The DESI+CMB joint fit measures

$$\left. \begin{aligned} w_0 &= -0.54^{+0.19}_{-0.21}, \\ w_a &= -1.39^{+0.61}_{-0.50}, \end{aligned} \right\} \begin{aligned} &\text{DESI BAO+Ly}\alpha \text{ FS} \\ &+\text{CMB} \end{aligned} \quad (31)$$

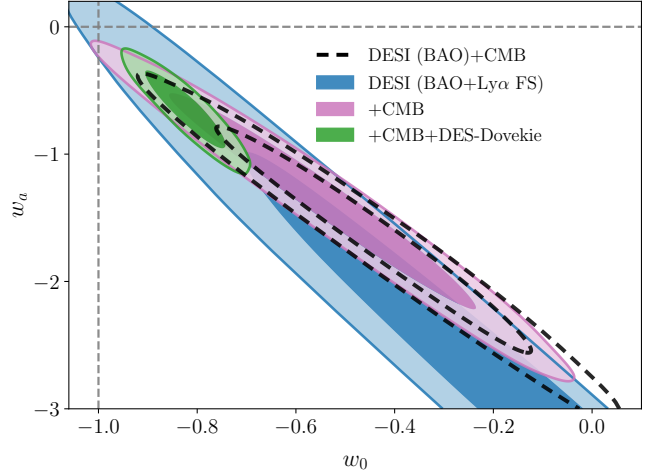


FIG. 10. 68% and 95% confidence contours on the dark energy parameters w_0 and w_a from DESI alone (blue) and in combination with other data sets. The intersection of the dashed gray lines represents the Λ CDM model. We show the previous DESI DR2 BAO constraints with the updated CMB combination used here in dashed black contours and its equivalent with the new DESI Ly α measurement in the purple contour. The updated DESI and CMB data show a preference of 2.7σ for w_0w_a CDM compared to Λ CDM. This preference increases to 3.2σ when also including SNe from the DES-Dovekie sample (green).

with a $\Delta\chi^2_{\text{MAP}} = -9.8$ relative to Λ CDM. This corresponds to a preference for evolving dark energy of 2.7σ . Therefore, adding the new Ly α constraints leads to a roughly $\sim 0.3\sigma$ shift towards Λ CDM when compared to the DESI (BAO)+CMB constraint, as illustrated in Figure 10 (dashed black versus pink contours). This is consistent with the updated AP measurement being in better agreement with *Planck* CMB, as well as the DESI posteriors on Λ CDM measuring a slightly higher Ω_m value. A high value of $\Omega_c h^2$ inferred by the CMB can also be correlated with the evidence for evolving dark energy from the DESI+CMB combination.

On the other hand, it has been pointed out that the value of the optical depth to reionization, τ , from the CMB is also correlated with the preference for evolving dark energy [131, 132]. A much higher value of the optical depth would effectively lead to a lower value of Ω_m , making DESI and the CMB consistent. We perform a test of this scenario within the Λ CDM model by dropping the low- ℓ EE CMB likelihood. This leads to $\Omega_m = 0.2998 \pm 0.0035$ and $\tau = 0.0905 \pm 0.0096$ for the combination of DESI and remaining CMB data. However, this is incompatible with the Ω_m value inferred by the SNe samples and of course inconsistent with the CMB low- ℓ E mode measurements by *Planck*. It would also raise the amplitude of clustering, which will be tested with forthcoming galaxy clustering and lensing analyses. Independently of this, high optical depth scenarios can be tested with astrophysical probes of reionization [e.g

133–138]. We compare the best-fit model from this test with the w_0w_a CDM fit and various other models in [Section VIE](#) below.

The inferred dark energy equation of state $w(z)$ still favors a phantom-crossing-like behaviour, indicating a $w(z) < -1$ during matter-domination and a $w(z) > -1$ at low redshift, where we only have one D_V/r_d measurement from the BGS tracer. Thus, to extend our analysis we use the DES-Dovekie SNe sample to provide constraints from the distance modulus and constrain the low-redshift information not fully accessible by BAO. In combination with DESI and CMB, we obtain

$$\left. \begin{aligned} w_0 &= -0.821 \pm 0.054, \\ w_a &= -0.65 \pm 0.20. \end{aligned} \right\} \begin{array}{l} \text{DESI BAO+Ly}\alpha \text{ FS} \\ \text{+CMB+DES-Dovekie} \end{array} \quad (32)$$

The associated preference for evolving dark energy under this combination is 3.2σ with $\Delta\chi^2_{\text{MAP}} = -12.9$. The use of alternative SNe datasets shows a significance of 3.0σ ($\Delta\chi^2_{\text{MAP}} = -11.5$) and 3.5σ ($\Delta\chi^2_{\text{MAP}} = -15.4$) for the updated Pantheon+ and Union3.1 samples, respectively. The complete set of constraints corresponding to these combinations and beyond is shown in [Table III](#).

We can also compute the pivot redshift z_p at which the dark energy equation of state, $w_p = w(z_p)$, is best constrained. While the pivot redshift depends on the dataset combination and we expect w_p to be close to -1 since the pivot redshift is close to the crossing redshift, this quantity can still give us an insight into the precision at which we are constraining the dark energy equation of state. In particular, we find that $w_p = -1.030 \pm 0.040$ with $z_p = 0.55$ for DESI+CMB, and $w_p = -0.981 \pm 0.022$ with $z_p = 0.32$ for DESI+CMB+DES-Dovekie. This represents roughly an 8% precision increase with respect to the values quoted in [\[5\]](#). Finally, we notice that the figure-of-merit (FoM) based on w_0 - w_a from DESI+CMB+DES-Dovekie reaches 227.4, which is 4% greater with respect to the FoM if we only use DESI BAO.

2. Neutrino masses

We also present updated constraints on the sum of neutrino masses. With the analysis of DR2 BAO, DESI already placed tight constraints on the sum of neutrino masses, assuming Λ CDM with three degenerate neutrino species and the standard physical prior that $\sum m_\nu > 0$ eV. In combination with CMB data from *Planck* PR4 and CMB lensing reconstruction from *Planck* and ACT DR6, DESI found $\sum m_\nu < 0.0642$ eV (95%) [\[5, 121\]](#). Compared to this analysis, we now additionally include new data from ground-based CMB experiments (see [Section VIA](#)). Using DESI DR2 BAO without Ly α FS, this yields a stronger upper limit of $\sum m_\nu < 0.0565$ eV (95%). However, with the addition of Ly α FS, this re-

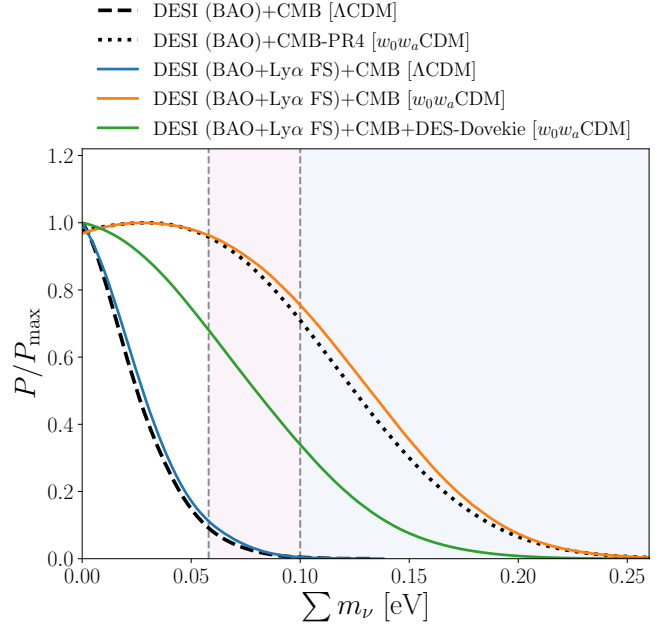


FIG. 11. Marginalized posterior distributions for the sum of neutrino masses, $\sum m_\nu$, from combinations of DESI, CMB, and SN data for Λ CDM and w_0w_a CDM. The constraints from DESI DR2 BAO without Ly α FS combined with CMB data are shown in black for both Λ CDM (dashed) and w_0w_a CDM (dotted). The addition of Ly α FS shifts the posteriors slightly towards larger masses (blue and orange). Including DES-Dovekie SN data tightens the w_0w_a CDM posterior (green), which remains compatible with the lower limits from neutrino oscillations [\[139\]](#).

laxes slightly to

$$\sum m_\nu < 0.0592 \text{ eV.} \quad \begin{array}{l} (\Lambda\text{CDM; 95\%): \\ \text{DESI BAO+Ly}\alpha \text{ FS} \\ \text{+CMB} \end{array} \quad (33)$$

This 2σ upper limit is nearly identical to the lower limit from neutrino oscillations under the normal mass ordering, $\sum m_\nu > 0.059$ eV [\[139\]](#). Moreover, the inverted ordering is excluded at more than 3σ . However, this should not be interpreted as strong evidence for the normal ordering, since most of the posterior volume is inconsistent with both mass orderings. Comparing the two cases, we find a modest $\Delta\chi^2_{\text{MAP}} = -5.2$ in favor of the normal ordering. We show the one-dimensional marginalized posterior distributions for various data combinations in [Figure 11](#).

As explained in [Section VIB](#), the upper limit is relaxed when we adopt a w_0w_a CDM background. Combining DESI BAO, Ly α FS, and CMB data, we find an upper limit of $\sum m_\nu < 0.166$ eV (95%). [Figure 11](#) demonstrates that the posterior distribution now peaks in the positive mass range and reveals the slight shift towards larger neutrino masses after the addition of the Ly α FS mea-

surement. Additionally including SN data tightens this limit again to

$$\sum m_\nu < 0.128 \text{ eV}, \quad (w_0 w_a \text{CDM}; 95\%): \quad \text{DESI BAO+Ly}\alpha \text{ FS} \quad (34) \\ + \text{CMB+DES-Dovekie}$$

which is very close to the result presented in [5]. The forthcoming full-shape analysis of galaxy and quasar clustering from DESI DR2 will yield even greater constraining power. Crucially, this will also bolster the robustness of the constraints by providing information from neutrino free streaming [121, 140].

3. Curvature

Cosmological distance measurements also allow us to test deviations from flatness within Λ CDM. The spatial curvature density parameter, Ω_K , can affect the measurements of Ω_m across the redshift tracers, and even impact measurements of neutrino masses, relaxing the neutrino mass bounds (see [141, 142], for instance). While cosmological data have been historically found to be consistent with a flat Λ CDM model, [143] have pointed out that a negative spatial curvature value (or equivalently, a positive Ω_K value) can alleviate some of the current cosmological tensions, with the high-redshift window probed by Ly α being key to set constraints that can distinguish between deviations from flatness, evolving dark energy, or the Λ CDM model. Therefore, here we evaluate the implications of the new Ly α FS measurement regarding spatial curvature.

In this scenario, CMB measurements are more susceptible to exhibiting geometrical degeneracies between Ω_m and Ω_K . Fortunately, this is partially mitigated by CMB lensing and the inclusion of DESI distance measurements which can help break the degeneracy. The updated DESI constraints on spatial curvature are

$$10^3 \Omega_K = 51 \pm 32, \quad \text{DESI BAO+Ly}\alpha \text{ FS} \quad (35)$$

where we find that the addition of Ly α FS leads to a 22% improvement in the Ω_K precision with respect to DESI BAO. When CMB data are included, the DESI+CMB combination yields

$$10^3 \Omega_K = 2.1 \pm 1.1, \quad \text{DESI BAO+Ly}\alpha \text{ FS} \quad (36) \\ + \text{CMB}$$

in agreement with previous BAO+CMB analyses [90, 92, 93]. The corresponding confidence contours for Ω_K and Ω_m are shown in Figure 12. We find a mild statistical preference for an open Universe, with the flat case lying

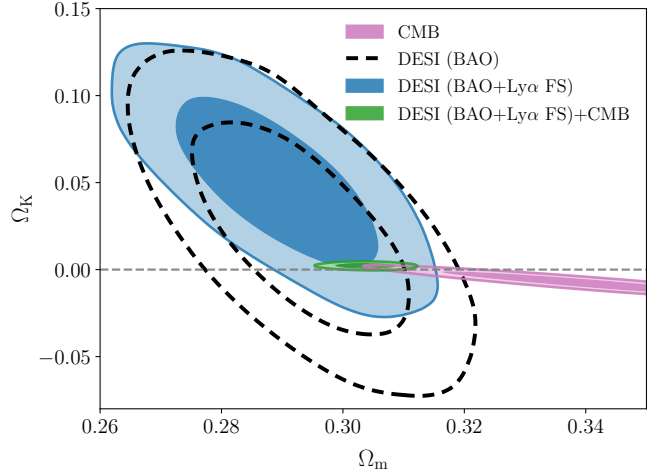


FIG. 12. Marginalized constraints in the Ω_m - Ω_K plane for a model with free curvature (i.e. Λ CDM + Ω_K). We show DESI results with and without the new Ly α forest constraints (blue versus dashed black). Both results are consistent with a flat Universe, and adding Ly α FS leads to a 22% improvement in the Ω_K precision. Combining DESI and CMB data breaks the degeneracy present in CMB constraints (pink), leading to a tight measurement of Ω_K that remains close to zero within $\sim 2\sigma$ (green).

approximately at the boundary of the 2σ confidence region. More precisely, a $\Delta\chi^2_{\text{MAP}} = -4.3$, corresponding to a 2.0σ preference over flatness.

Finally, the inclusion of the DES-Dovekie SNe data has a minor impact on the curvature constraints, leading to $10^3 \Omega_K = 2.2 \pm 1.0$. As expected, the inclusion of SNe data does not lead to a noticeable improvement, since the geometric Ω_K - Ω_m degeneracy has already been largely broken by the BAO+CMB combination, and DESI provides higher precision on curvature constraints since it is able to access higher redshifts.

Overall, despite the mild 2σ fluctuation toward positive curvature values, the constraints are consistent with flatness, and upcoming DESI data releases will help to further constrain spatial curvature.

E. Summary and discussion

We now summarize the results found across the different models. The best-fit to the DESI data is presented for each model in Figure 13, where we use the *Planck* Λ CDM model as reference and fit all other models to DESI+CMB. Overall, the new D_M/D_H Ly α FS measurement at $z_{\text{eff}} = 2.33$ is consistent with the *Planck* Λ CDM model within 0.5σ . For extensions, this $z_{\text{eff}} = 2.33$ measurement is consistent with models such as DESI+CMB Λ CDM, with or without spatial curvature, to within 1.1σ . This measurement, however, does not favor evolving dark energy, with the best-fit for $w_0 w_a$ CDM being around 2σ apart from the Ly α FS data point. We summarize the

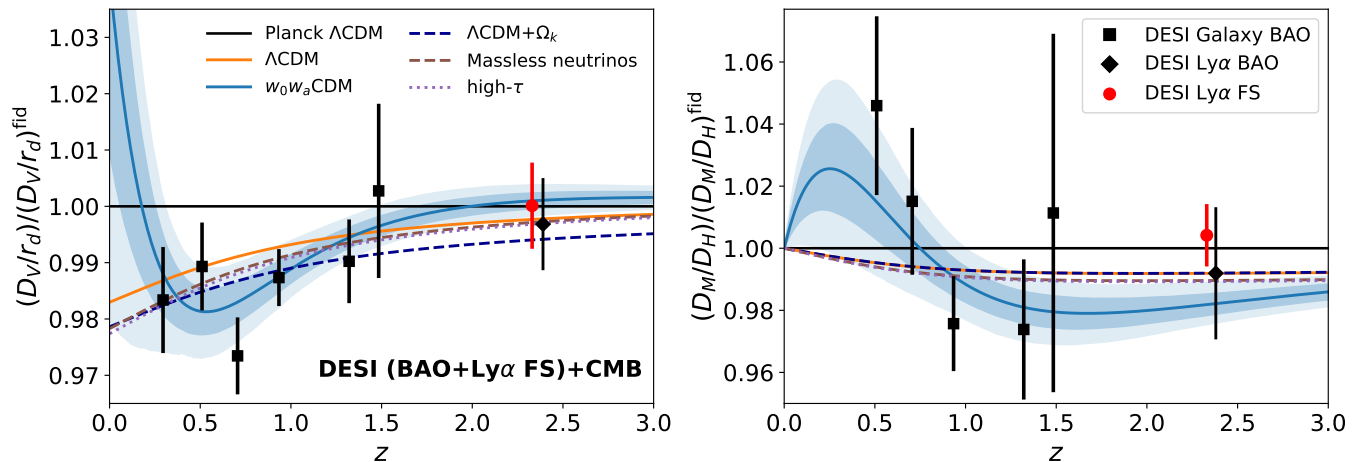


FIG. 13. Measurements of cosmological distances, D_V in r_d units and the ratio D_M/D_H , relative to the fiducial *Planck* Λ CDM model (horizontal black line). We show the DESI DR2 BAO measurements and the new DESI DR2 Ly α FS measurement. The DESI + CMB best-fit w_0w_a CDM model that provides the highest improvement with respect to Λ CDM is shown in blue, with the color bands representing the 68% and 95% confidence regions. We also show other DESI + CMB best-fit models tested in this work as different colored lines. These models include Λ CDM (orange), free curvature (dashed blue), massless neutrinos (dashed brown), and the high- τ scenario (dotted), which is a Λ CDM fit where we drop the low- ℓ EE CMB likelihood.

improvements of the fit with respect to flat Λ CDM for various data combinations and extensions considered in this work in Table IV.

Together with the new Ly α FS measurement, the inclusion of CMB data anchors the high-redshift fits to be close to *Planck* Λ CDM. However, as we approach lower redshifts, the values of the spherically-averaged distance, D_V/r_d , are systematically low with respect to *Planck* Λ CDM. Such a deficit in the distances measured by BAO can be explained by a variety of models, with w_0w_a CDM mildly outstanding since it has the flexibility to better fit the LRG2 BAO measurement. Models such as Λ CDM+ Ω_K , Λ CDM with three massless neutrinos instead of the baseline neutrino with a mass of 0.06 eV, and allowing for a high value of the optical depth τ (by excluding low- ℓ CMB polarization data) provide quite similar fits in D_V/r_d and particularly D_M/D_H . However, with the exception of w_0w_a CDM, none of these models are able to capture the data points in the AP measurements below $z < 1$. The upcoming DESI full-shape analysis from galaxy and quasar clustering at these redshifts may shed further light on this feature, while the DR3 BAO analysis could provide further insights into the AP measurement at $z_{\text{eff}} = 2.33$.

VII. CONCLUSIONS

We have presented a measurement of the Alcock–Paczynski (AP) effect using the full shape of Ly α forest correlation functions measured from the second DESI data release (DR2). Our data set includes over 820,000 Ly α forest spectra and the positions of over 1.2 million quasars, which are used to compute the Ly α -QSO cross-

Datasets	$\Delta\chi^2_{\text{MAP}}$	Significance
w_0w_aCDM		
DESI	−5.6	1.9 σ
DESI+CMB	−9.8	2.7 σ
DESI+DES-Dovekie	−6.7	2.1 σ
DESI+CMB+DES-Dovekie	−12.9	3.2 σ
ΛCDM + Ω_K		
DESI	−2.4	1.5 σ
DESI+CMB	−4.3	2.0 σ
DESI+DES-Dovekie	−4.3	2.0 σ
DESI+CMB+DES-Dovekie	−4.8	2.2 σ

TABLE IV. Summary of the effective χ^2_{MAP} differences compared to Λ CDM for different models and datasets. The differences are listed for the best-fit w_0w_a CDM model relative to a Λ CDM model with $w_0 = -1$ and $w_a = 0$, and similarly for the best-fit model with spatial curvature compared to flat Λ CDM. The corresponding frequentist significance for the preference over flat Λ CDM is listed in the third column.

correlation. Our combined AP measurement corresponds to a 1% constraint on the ratio D_M/D_H at an effective redshift $z_{\text{eff}} = 2.33$, where D_M is the comoving angular diameter distance and D_H is the ratio of the speed of light and the Hubble parameter at z_{eff} . This is the tightest constraint on D_M/D_H at redshifts $z > 1$ so far.

Our analysis is based on a template fit of the Ly α auto-correlation and its cross-correlation with quasars. Fol-

lowing [43], our template is split into a peak component and a smooth component, similar to the approach used in BAO analyses. This allows us to separate the AP information from the broadband and the BAO peak, which means our analysis builds on the DESI DR2 BAO analysis by also extracting AP information from the broadband of Ly α forest correlations (Section III A). Relative to previous Ly α full-shape and BAO analyses, we have also added two major improvements to our model for Ly α forest correlations. These include a framework for marginalizing over small-scale information that is spread to large scales due to the quasar continuum fitting procedure (Section III C), and a model for a scale-dependent Ly α forest bias arising from larger-scale fluctuations in the UV background (Section III D).

We have validated our analysis with two types of mocks and with a wide range of tests on blinded data, which are presented in detail in the companion paper H26. For our validation, we use a threshold of one-third of the statistical uncertainty to determine whether a test passes or fails, in line with previous DESI Ly α analyses [13, 14]. The two types of mocks we use include Ly α forest mocks from the *AbacusSummit* simulations [67] and a new set of mocks based on second-order Lagrangian Perturbation Theory (2LPT), which are presented in a companion paper [28]. Our model configuration and scale cuts are based on tests with these two types of mocks as described in Section IV A. We target large scales in the Ly α forest correlation functions, where a linear-theory approach provides a good fit and unbiased constraints. The tests with mocks show that our baseline analysis recovers unbiased AP constraints within our threshold (Section IV A). Finally, we also present BAO results from the new mocks, which led us to update the systematic error budget on BAO with respect to the one used in the DESI DR2 BAO analysis [14] to include a slightly larger uncertainty in the AP direction (Section V B).

In Section IV B and the companion work H26, we present the validation with blinded data. This includes several data splits and a large set of analysis variations designed to test the robustness of our measurement to different analysis and modeling choices. All data splits are consistent with each other, and we do not find significant outliers in our set of analysis variations. In particular, modeling variations show consistent results for various choices of how contaminants and small-scale nonlinearities are treated, for more conservative scale cuts, and when using different priors. Based on these results, and the validation with mocks, we conclude that our AP result is robust (Section IV C). The validation process was performed on blinded data, and the baseline analysis choices were decided before unblinding the measurement, with no further changes after unblinding.

Our Ly α forest results are presented in Section V A, where we show the model fits to the Ly α correlation functions and the broadband AP and BAO results. The full-shape model provides a good fit to the correlation functions, and we find that AP constraints from the broad-

band are consistent with those from the BAO peak. Our new AP constraint is a factor of 2 tighter compared to the BAO measurement from the same data set [14]. We use these constraints, along with the extra systematic uncertainty on BAO (Section V B), to infer the corresponding cosmological distances in Section V C. We present the D_M/D_H constraint from our broadband AP measurement, along with the combined broadband AP and BAO constraints on the ratios $D_M(z_{\text{eff}})/r_d$ and $D_H(z_{\text{eff}})/r_d$, where r_d is the size of the acoustic scale at the drag epoch and $z_{\text{eff}} = 2.33$. Our measurements correspond to 0.84% and 0.77% constraints on D_M/r_d and D_H/r_d , respectively. The precision of these results exceeds the requirements of the full DESI survey in the redshift range $1.9 < z < 4.0$ [2, 19], with just three years of data.

We present the cosmological interpretation of our measurement alone and in combination with DESI BAO measurements at lower redshifts and other external data sets in Section VI. The Ly α full-shape (FS) measurements are consistent with both DESI BAO and CMB predictions within Λ CDM. Furthermore, the inferred matter fraction, Ω_m , constraint from the Ly α forest is consistent with SNe results. Combining our measurement with DESI BAO constraints from galaxy clustering leads to an $\sim 8\%$ improvement with respect to DESI DR2 BAO (which includes Ly α BAO), and a small shift to larger values of Ω_m . This leads to a decrease in the tension between DESI and the CMB from 2.4σ to 2.2σ . We also combine our Ly α forest measurement with a BBN prior on the baryon density in order to calibrate the acoustic scale. This results in a 2% constraint on the Hubble constant, $H_0 = 66.5 \pm 1.3$, entirely from high-redshift data (Ly α +BBN) and independent of CMB anisotropies.

Section VI D presents our constraints on extended models beyond Λ CDM from the DESI BAO+Ly α FS combination. When considering models with an evolving dark energy equation of state parametrized by w_0 and w_a , the new Ly α forest AP measurement is in $\sim 2\sigma$ tension with the w_0w_a CDM prediction of AP at $z = 2.33$ for both DESI+CMB and DESI+CMB+SNe. Therefore, adding Ly α FS leads to a small decrease in the preference for w_0w_a CDM relative to Λ CDM when combined with other data sets. For DESI+CMB, the preference decreases from 3.2σ to 2.7σ , while for DESI+CMB+SNe, it has a milder impact, decreasing the tension from 3.4σ to 3.1σ . We also present updated results on the sum of the neutrino masses, $\sum m_\nu$, where we find that the Ly α result leads to a very mild relaxation of the upper bound on $\sum m_\nu$, due to its preference for a higher Ω_m value. For models with free curvature, Ω_K , we find that DESI BAO+Ly α FS is consistent with a flat Universe within 1.5σ , while the DESI+CMB constraint is 2σ away from a flat Universe, in line with [5]. Our cosmological results are compared and discussed in Section VI E.

Ly α forest measurements provide the opportunity to test cosmological models in a unique regime at $z > 2$, which has so far been difficult to access with large galaxy surveys. Our measurements come from a time when the

Universe was still deep in the matter domination era, providing an anchor to low-redshift results for constraining the expansion history and the evolution of dark energy, and to the CMB for constraining physics during matter domination (Section VIB). The DESI DR2 results allow the Ly α forest to begin to separate at the $\sim 2\sigma$ level between state-of-the-art dark energy constraints for the first time. These measurements will continue to improve with future DESI data releases, and future methodology improvements may also allow us to access information on smaller scales leading to even tighter constraints.

VIII. DATA AVAILABILITY

The data used in this analysis will be made public along with Data Release 2 (details in <https://data.desi.lbl.gov/doc/releases/>). The data points corresponding to the figures from this paper will be available in a Zenodo repository.

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The authors are honored to be permitted to conduct scientific research on I’oligam Du’ag (Kitt Peak), a mountain with particular significance to the Tohono O’odham Nation.

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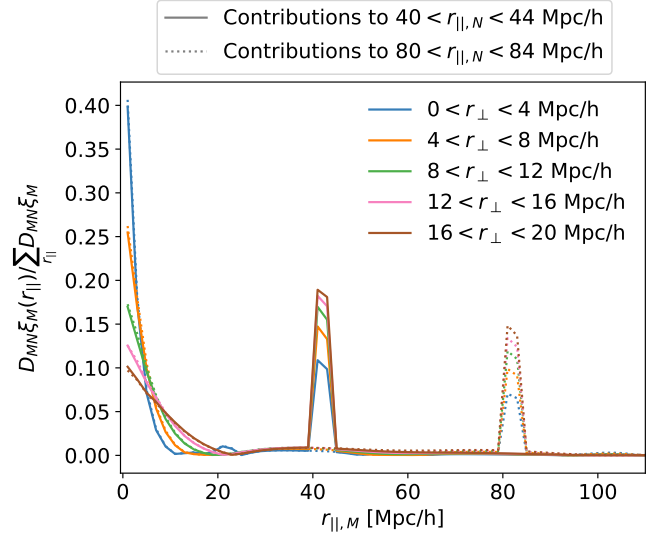


FIG. 14. Fractional contributions to one bin of the distorted model ξ_N from one transverse bin in the undistorted model ξ_M as a function of the line-of-sight separation in the undistorted space, $r_{||,M}$. Different colors display different choices of the transverse bin, which is the same for both ξ_M and ξ_N . We show contributions to two different distorted $r_{||,N}$ bins using continuous and dotted lines, respectively.

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Appendix A: Further details on SSM

In this work, we introduced a new method for small-scale marginalization (SSM) of distorted Ly α forest correlation functions, as described in [Section III C](#). Here, we wish to provide more details on the motivation and consequences of this new method.

We begin with the motivation for introducing the new

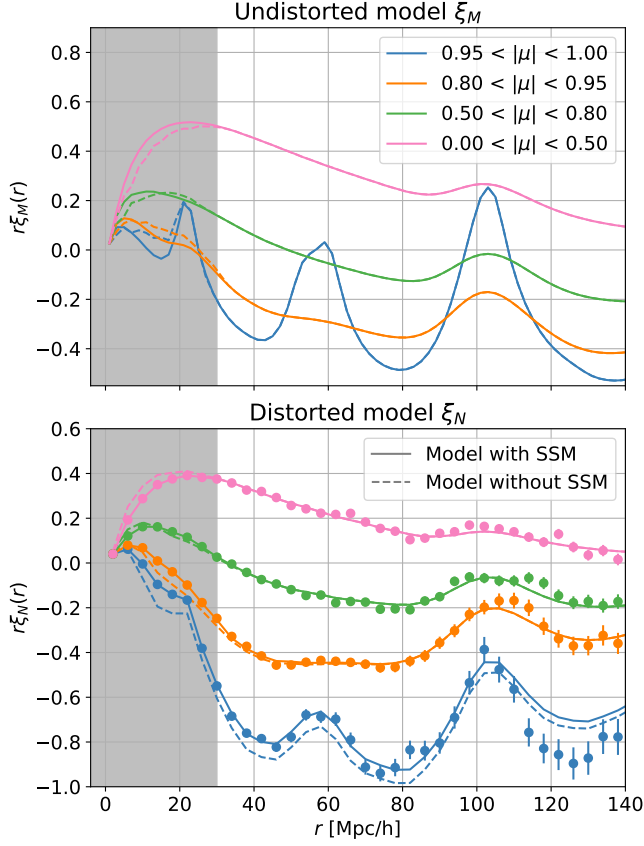


FIG. 15. Best-fit model compressed into wedges as a function of μ and plotted in the undistorted space (top) and in the distorted space (bottom). We show the same model with and without the small-scale marginalization (SSM) terms as continuous and dashed lines, respectively. The data is also shown in the bottom panel.

SSM method. As shown in Equations (13) to (15), we use a distortion matrix that multiplies our undistorted model ξ_M , such that the resulting distorted model ξ_N can then be fitted to the data. The distortion matrix is built from the projection matrix of each forest in Equation (2), and results in a mixing of the information along the line-of-sight. We showcase this mixing in Figure 14, which displays the fractional contributions from each undistorted line-of-sight bin $r_{||,M}$, to a given distorted line-of-sight bin $r_{||,N}$, for a fixed transverse bin $r_{\perp} = r_{\perp,M} = r_{\perp,N}$. Contributions to the $40 < r_{||,N} < 44h^{-1}$ Mpc bin are shown using continuous lines, while those to the $80 < r_{||,N} < 84h^{-1}$ Mpc bin are shown using dotted lines. Different colors indicate different choices of $r_{\perp}$.

Figure 14 shows that for each distorted bin $r_{||,N}$, a significant amount of information comes from small scales in the undistorted correlation $r_{||,M} < 20h^{-1}$ Mpc. For the first $r_{\perp}$ bin shown in blue ($0 < r_{\perp} < 4h^{-1}$ Mpc), about 70% of the information comes from the small scales, and only $\sim 20\%$ comes from $r_{||,M} = r_{||,N}$. By the fifth $r_{\perp}$

bin shown in brown ($0 < r_{\perp} < 4h^{-1}$ Mpc), the contribution from small scales drops to $\sim 45\%$ while that from $r_{||,M} = r_{||,N}$ increases to $\sim 40\%$. On the other hand, comparing the dotted and continuous lines of the same color shows that the contribution from small scales is almost identical for very different choices of $r_{||,N}$. While not displayed in Figure 14, when fitting small scales in ξ_N , the vast majority of the information, between 95% and 99% for $r_{\min} = 30h^{-1}$ Mpc, comes from those same scales in ξ_M (i.e., from $r < r_{\min}$).

This motivates our approach of introducing one free parameter to marginalize over each small-scale bin in ξ_M , while fitting the corresponding ξ_N bins to the real data. Each small-scale ($r_{||,M}, r_{\perp,M}$) bin is marginalized over, and has a corresponding fitted ($r_{||,N}, r_{\perp,N}$) data bin. This ensures that only the large scales in our model are used to fit the data, and that we do not lose information due to the large number of extra degrees of freedom. Our method can be thought of as using the distortion matrix to deconvolve the data, and then applying scale cuts to the deconvolved data.

We show the impact of SSM on our best-fit model in Figure 15. The continuous and dashed lines represent the same model, with and without the SSM terms.²⁵ The top panel shows the undistorted model, where the marginalized terms only affect scales $r < r_{\min}$ (displayed using the gray band). The bottom panel shows the distorted model and the data. In this case, the marginalized small-scale terms have an impact on all scales, especially in the line-of-sight wedge (blue). Furthermore, due to how our method is set up, the $r_N < r_{\min}$ data bins are perfectly fitted by our model with SSM.

We have validated the new SSM method using mocks, as discussed in Section IV A and H26. In the case of contaminated CoLoRe 2LPT mocks, we found that using SSM does not lead to a significant difference in AP constraints for $r_{\min} \geq 30h^{-1}$ Mpc. On the other hand, for AbacusSummit we found percent-level biases in AP, which disappear when using SSM, as seen in Figure 2.

Finally, we note that the SSM method relies on a choice of prior for the marginalization (the diagonal elements of the $\mathbf{S}$ matrix in Equation (16)). For numerical stability, we use a finite value $\mathbf{S}_{ii} = 100$, which is significantly larger than the amplitude of the model bins that are marginalized (by about 4 orders of magnitude). We have tested that changing this value by a factor of 2 in either direction does not impact our results.

Appendix B: Nuisance parameters

Our model has a total of 21 free parameters, excluding those fitting small-scale bins which are analytically

²⁵ This means that the model without SSM is not re-fitted to the data, which is why it is not a good fit to the line-of-sight wedge in the bottom panel.

Parameter	Priors	Results
α_p	$\mathcal{U}[0.50, 1.50]$	1.0007 ± 0.0061
ϕ_p	$\mathcal{U}[0.50, 1.50]$	1.001 ± 0.020
ϕ_s	$\mathcal{U}[0.50, 1.50]$	1.007 ± 0.011
α_s	$\mathcal{U}[0.50, 1.50]$	1.044 ± 0.036
f	$\mathcal{U}[0.00, 2.00]$	$1.43^{+0.17}_{-0.20}$
b_F	$\mathcal{U}[-1.00, 0.00]$	$-0.167^{+0.010}_{-0.015}$
β_F	$\mathcal{U}[0.00, 3.00]$	$1.193^{+0.083}_{-0.12}$
b_Q	$\mathcal{U}[0.00, 6.00]$	3.49 ± 0.22
σ_z [Mpc/h]	$\mathcal{U}[0.00, 15.00]$	4.8 ± 1.6
$\Delta r_{ }$ [Mpc/h]	$\mathcal{N}(0.0, 1.0)$	0.34 ± 0.26
ξ_0^{TP}	$\mathcal{U}[0.00, 2.00]$	< 0.24
b_{HCD}	$\mathcal{U}[-0.50, 0.00]$	> -0.027
β_{HCD}	$\mathcal{N}(0.500, 0.090)$	0.498 ± 0.089
L_{HCD} [Mpc/h]	$\mathcal{N}(5.0, 2.0)$	4.9 ± 1.8
$10^3 b_{\text{SiII}(1190)}$	$\mathcal{U}[-500.00, 0.00]$	$-3.62^{+0.44}_{-0.38}$
$10^3 b_{\text{SiII}(1193)}$	$\mathcal{U}[-500.00, 0.00]$	$-3.12^{+0.42}_{-0.36}$
$10^3 b_{\text{SiII}(1207)}$	$\mathcal{U}[-500.00, 0.00]$	-9.50 ± 2.1
$10^3 b_{\text{SiII}(1260)}$	$\mathcal{U}[-500.00, 0.00]$	$-3.05^{+0.43}_{-0.38}$
$10^3 b_{\text{CIV}(\text{eff})}$	$\mathcal{N}(-19.0, 5.0)$	-19.80 ± 5.1
$10^4 a_{\text{noise}}$	$\mathcal{U}[0.00, 100.00]$	2.24 ± 0.2
b_γ	$\mathcal{U}[-1.00, 1.00]$	0.143 ± 0.046

TABLE V. Priors and parameter constraints (posterior mean and 68% credible intervals) for the full set of parameters, including the main parameters used for cosmology (first three) and the nuisance parameters.

marginalized (Section III C). We present the full set of priors and parameter constraints in Table V. We denote uniform priors as $\mathcal{U}[\text{min}, \text{max}]$, and Gaussian priors with mean μ and standard deviation σ as $\mathcal{N}[\mu, \sigma]$. The choice of priors closely follows that of the DR2 BAO analysis [14] and the DR1 Ly α full-shape analysis [16]. The impact of the choice of Gaussian priors has been tested in the companion work H26, and is discussed in Section IV C.

Several parameters have fairly different values compared to those obtained in the DR2 BAO analysis. This is because of a few different factors, including the change in

scale cuts, the improved model, and the fact that we vary the scale parameters of the smooth component (which are kept fixed in BAO analyses). Finally, we note that nuisance parameter results are presented for completeness purposes, and advise against interpretation of these values in other contexts. This is because we have not validated their constraints, and throughout the analysis we simply treated them as parameters to be marginalized over.

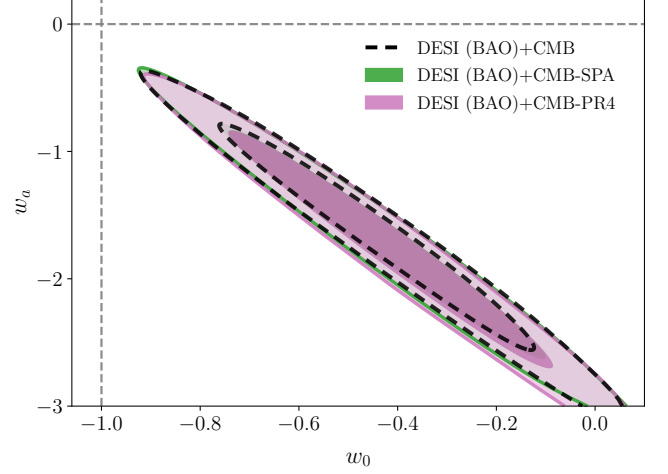


FIG. 16. Constraints on the dark energy equation of state parameters, w_0 and w_a , from DESI BAO (without Ly α FS) in combination with different CMB datasets. The contours represent the 68% and 95% credible intervals. The CMB-PR4 combination represents the likelihood choice from [5], the CMB-SPA combination is used in [93], and the CMB combination is the baseline choice described in Section VI A.

Appendix C: Updates on the CMB likelihood

Here, we provide a comparison of our baseline CMB combination with respect to other relevant CMB choices adopted in recent cosmological analyses. Figure 16 shows the comparison of constraints on $w_0 w_a$ CDM when DESI BAO is combined with CMB. We opt not to include the Ly α FS data here for a direct comparison to [5]. With DESI BAO and CMB data, the preference for evolving dark energy is around 3σ with minor 0.1-0.2 σ variations depending on the choice of CMB dataset.